wazuh.

Integration of Wazuh with pfSense Firewall

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Introduction

pfSense is an open-source firewall and router platform based on FreeBSD. It is widely recognized for its reliability, flexibility, and enterprise-grade features while remaining cost-effective. pfSense can be deployed on physical hardware, virtual machines, or as a cloud appliance.

It provides powerful functionalities such as:

- Firewall and Router Capabilities – Advanced traffic filtering, stateful packet inspection, and secure routing.
- VPN Services – Built-in support for IPSec, OpenVPN, and WireGuard to ensure secure remote access.
- Intrusion Detection and Prevention – Integration with IDS/IPS tools like Snort and Suricata.
- Traffic Shaping and QoS – Efficient bandwidth management for optimized network performance.
- High Availability and Load Balancing – Redundancy and failover features for enterprise continuity.

pfSense is commonly used in businesses, educational institutions, and even home networks, offering an affordable yet professional-grade solution for network security and management.

Achievements of Integrating pfSense with Wazuh

By integrating **pfSense** with **Wazuh**, we achieved a comprehensive and centralized security monitoring solution. This integration allows pfSense firewall logs to be forwarded to Wazuh, where they are analyzed for security threats, anomalies, and policy violations in real time.

Key achievements include:

- **Centralized Log Management** – All pfSense firewall events are collected and stored in Wazuh, making it easier to analyze and correlate security incidents.

- **Enhanced Threat Detection** – Wazuh applies its built-in rules to detect suspicious activities such as port scanning, brute-force attempts, and unauthorized access.

- **Real-Time Monitoring** – Security teams gain instant visibility into firewall activity through the Wazuh dashboard and alerts.

- **Incident Response Automation** – Automated notifications and responses reduce the time needed to contain threats.

- **Regulatory Compliance Support** – Helps meet compliance requirements (e.g., PCI DSS, GDPR, HIPAA) by ensuring proper log collection and reporting.

Step by Step Pfsense Installation Process

In this step, you need to download the ISO image file of pfSense.

Navigate to the official Netgate mirror and download the latest stable release:

https://atxfiles.netgate.com/mirror/downloads/

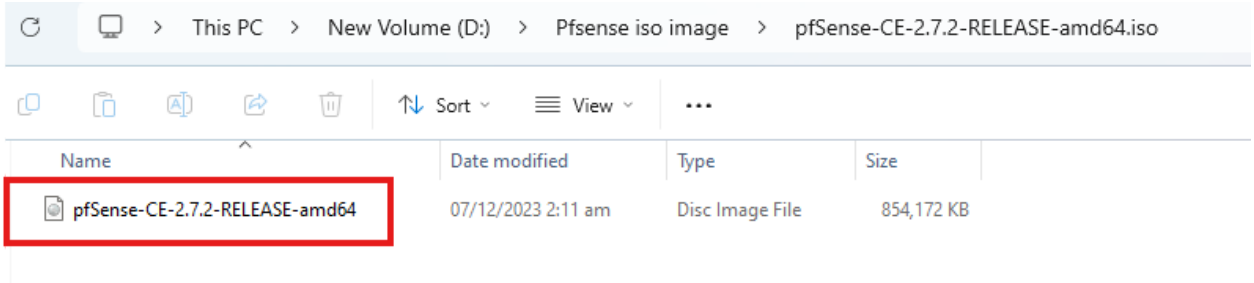
Index of /mirror/downloads/

../		
old/	06-Jun-2024 19:18	-
pfSense-CE-2.6.0-RELEASE-amd64.iso.gz	31-Jan-2022 20:31	437073513
pfSense-CE-2.6.0-RELEASE-amd64.iso.gz.sha256	31-Jan-2022 20:32	114
pfSense-CE-2.7.0-RELEASE-amd64.iso.gz	29-Jun-2023 20:11	495733706
pfSense-CE-2.7.0-RELEASE-amd64.iso.gz.sha256	29-Jun-2023 20:11	114
pfSense-CE-2.7.1-RELEASE-amd64.iso.gz	17-Nov-2023 00:47	574639430
pfSense-CE-2.7.1-RELEASE-amd64.iso.gz.sha256	17-Nov-2023 00:47	114
pfSense-CE-2.7.2-RELEASE-amd64.iso.gz	08-Dec-2023 18:27	574277009
pfSense-CE-2.7.2-RELEASE-amd64.iso.gz.sha256	08-Dec-2023 18:27	114
pfSense-CE-memstick-2.6.0-RELEASE-amd64.img.gz	31-Jan-2022 20:40	438161574
pfSense-CE-memstick-2.6.0-RELEASE-amd64.img.gz...>	31-Jan-2022 20:40	123
pfSense-CE-memstick-2.7.0-RELEASE-amd64.img.gz	29-Jun-2023 20:11	499043832
pfSense-CE-memstick-2.7.0-RELEASE-amd64.img.gz...>	29-Jun-2023 20:11	123
pfSense-CE-memstick-2.7.1-RELEASE-amd64.img.gz	17-Nov-2023 00:47	575598913
pfSense-CE-memstick-2.7.1-RELEASE-amd64.img.gz...>	17-Nov-2023 00:47	123
pfSense-CE-memstick-2.7.2-RELEASE-amd64.img.gz	08-Dec-2023 18:27	576633377
pfSense-CE-memstick-2.7.2-RELEASE-amd64.img.gz...>	08-Dec-2023 18:27	123
pfSense-CE-memstick-ADI-2.6.0-RELEASE-amd64.img.gz	31-Jan-2022 20:48	440116418
pfSense-CE-memstick-ADI-2.6.0-RELEASE-amd64.img..>	31-Jan-2022 20:49	127
pfSense-CE-memstick-ADI-2.7.0-RELEASE-amd64.img.gz	29-Jun-2023 20:11	496977920
pfSense-CE-memstick-ADI-2.7.0-RELEASE-amd64.img..>	29-Jun-2023 20:11	127
pfSense-CE-memstick-ADI-2.7.1-RELEASE-amd64.img.gz	17-Nov-2023 00:47	576330254
pfSense-CE-memstick-ADI-2.7.1-RELEASE-amd64.img..>	17-Nov-2023 00:47	127
pfSense-CE-memstick-ADI-2.7.2-RELEASE-amd64.img.gz	08-Dec-2023 18:27	575785298
pfSense-CE-memstick-ADI-2.7.2-RELEASE-amd64.img..>	08-Dec-2023 18:27	127
pfSense-CE-memstick-serial-2.6.0-RELEASE-amd64...>	31-Jan-2022 20:57	438946731
pfSense-CE-memstick-serial-2.6.0-RELEASE-amd64...>	31-Jan-2022 20:57	130
pfSense-CE-memstick-serial-2.7.0-RELEASE-amd64...>	29-Jun-2023 20:11	496954115
pfSense-CE-memstick-serial-2.7.0-RELEASE-amd64...>	29-Jun-2023 20:11	130
pfSense-CE-memstick-serial-2.7.1-RELEASE-amd64...>	17-Nov-2023 00:47	575162207
pfSense-CE-memstick-serial-2.7.1-RELEASE-amd64...>	17-Nov-2023 00:47	130
pfSense-CE-memstick-serial-2.7.2-RELEASE-amd64...>	08-Dec-2023 18:27	576488704
pfSense-CE-memstick-serial-2.7.2-RELEASE-amd64...>	08-Dec-2023 18:27	130

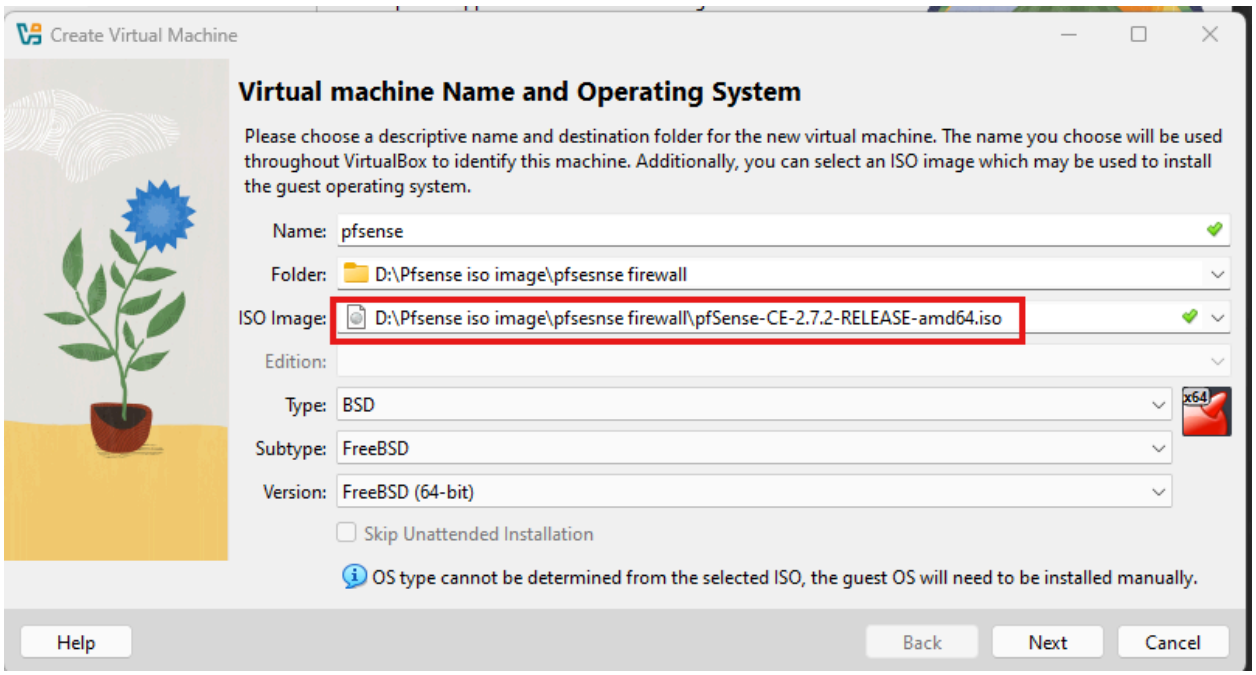
`pfSense-CE-2.7.2-RELEASE-amd64.iso.gz`

Once the download is complete, extract the compressed `.gz` file using a tool such as 7-Zip, WinRAR, or the built-in extraction utility of your operating system.
After extraction, you will obtain the installer file in the standard ISO format:
`pfSense-CE-2.7.2-RELEASE-amd64.iso`

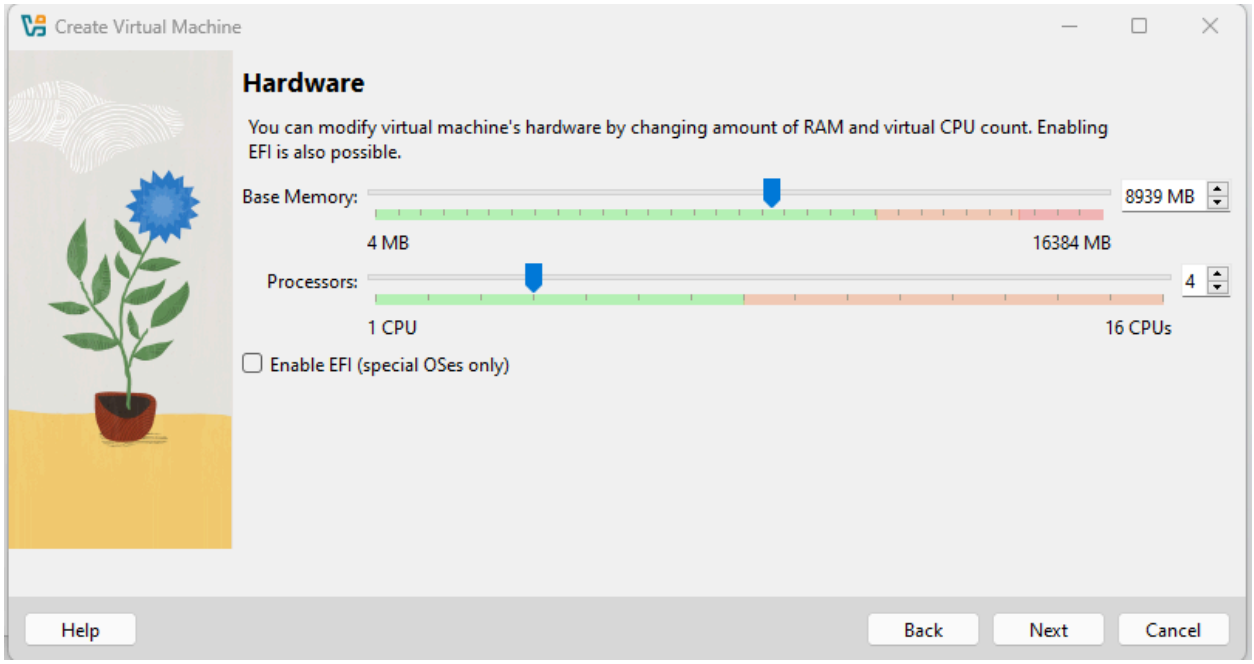
I successfully download the iso file.



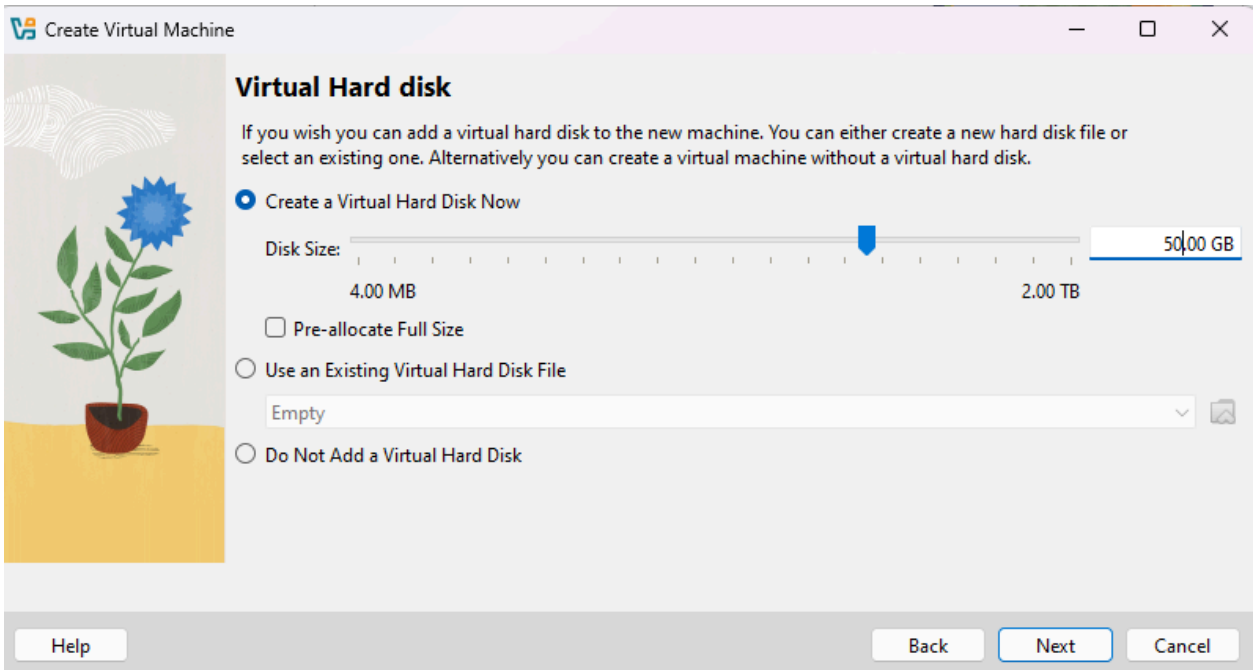
Open VirtualBox and begin creating a new virtual machine for pfSense. Assign an appropriate name to the VM and attach the pfSense ISO file as the installation media



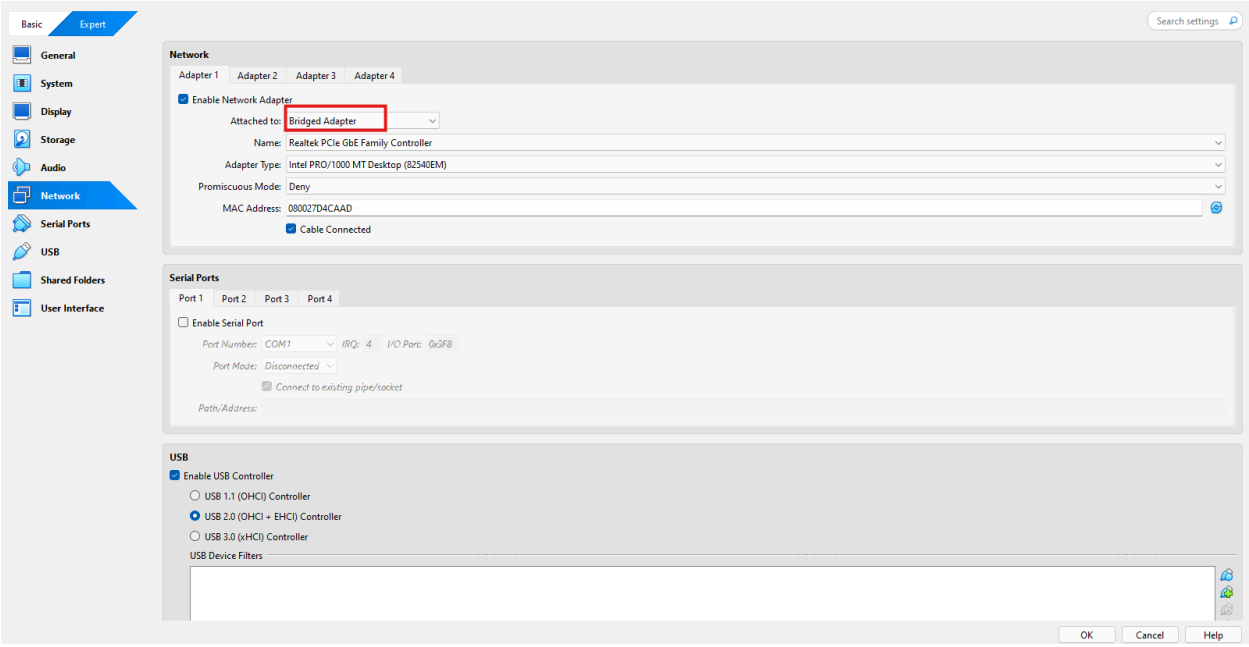
Configure the VM hardware by assigning RAM, CPU cores, a virtual hard disk, and the required network adapters.



Create a new dynamically allocated virtual hard disk of about 10–20 GB for the pfSense VM.

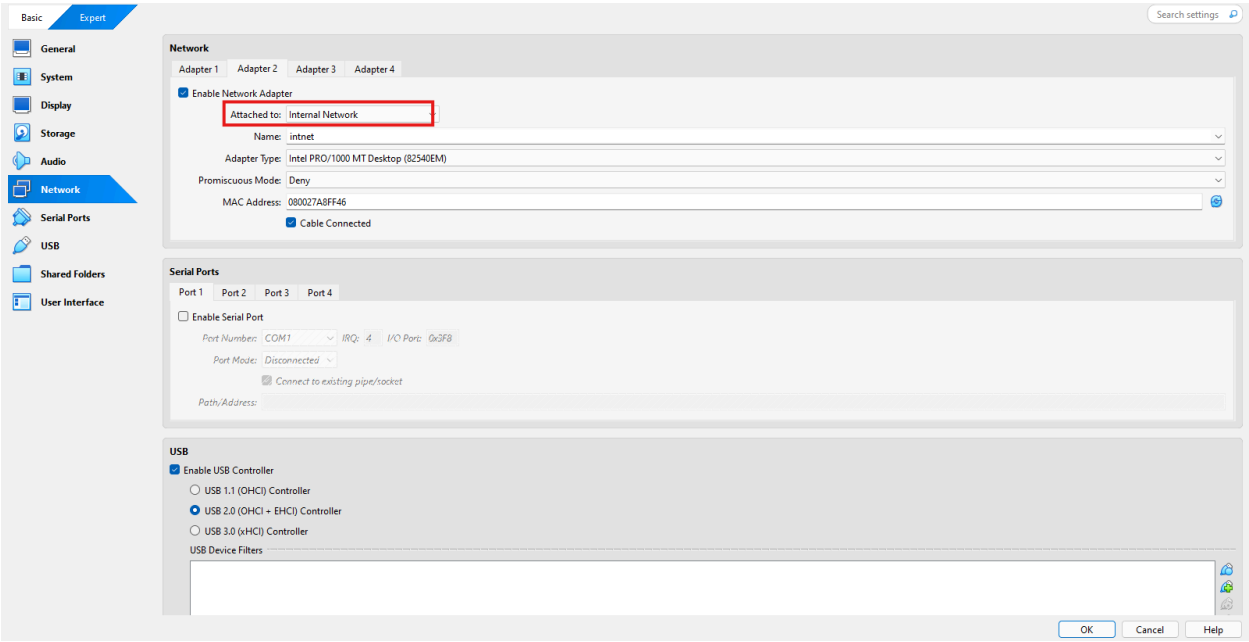


Next, configure Network Adapter 1 according to the settings shown in the screenshot.

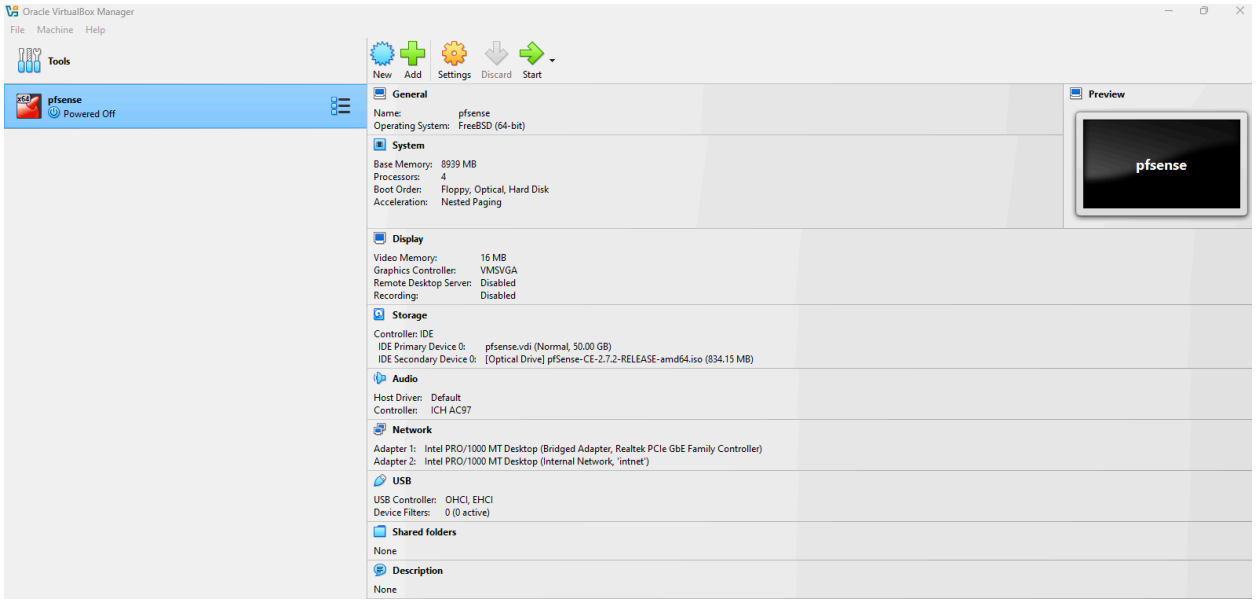


Enable Network Adapter 2 for the pfSense VM.

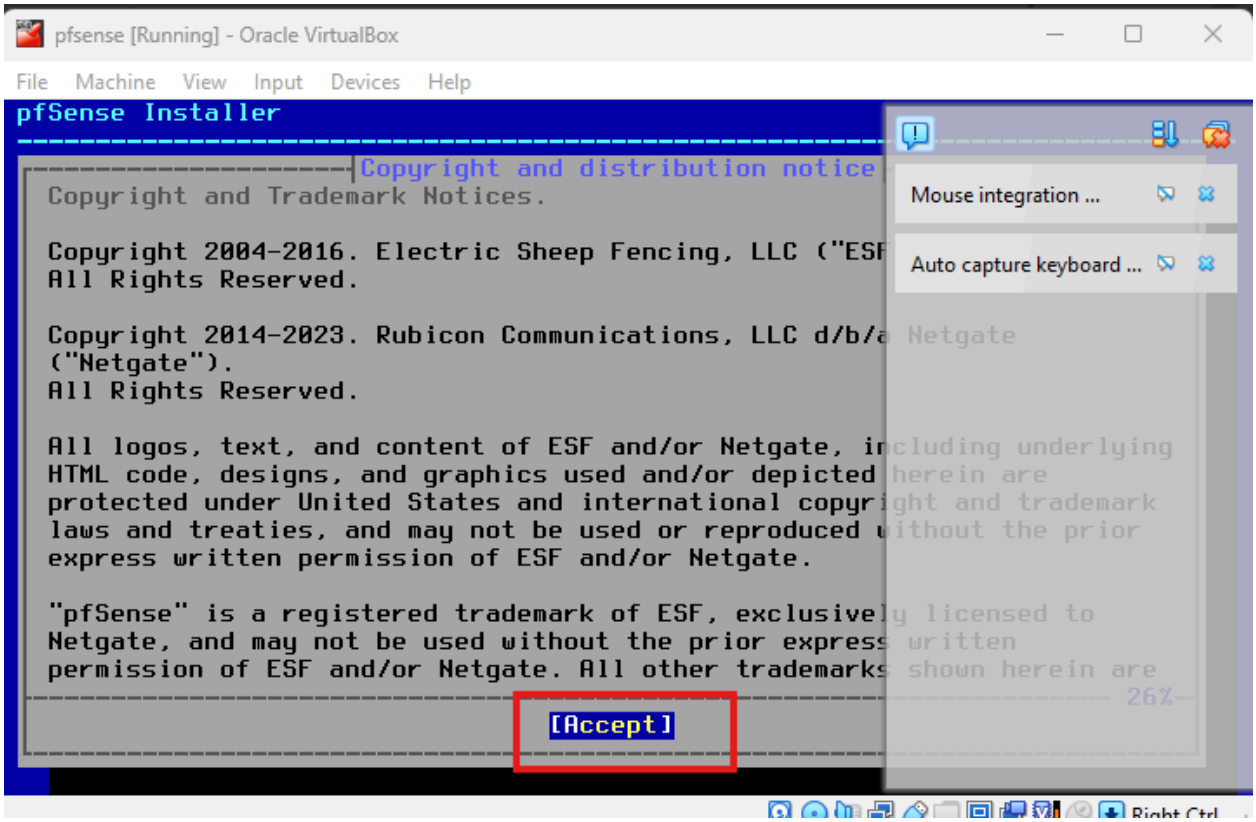
Configure Network Adapter 2 with the same settings shown in the screenshot.

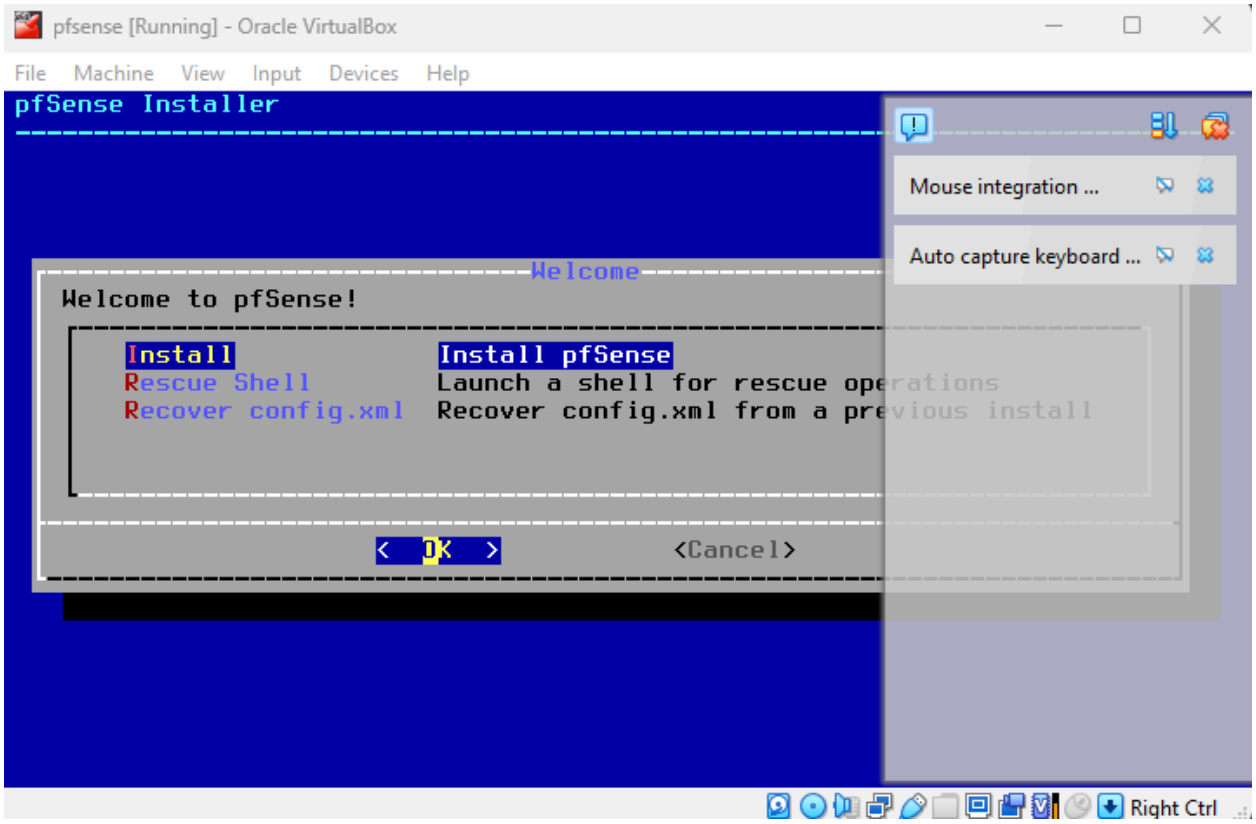


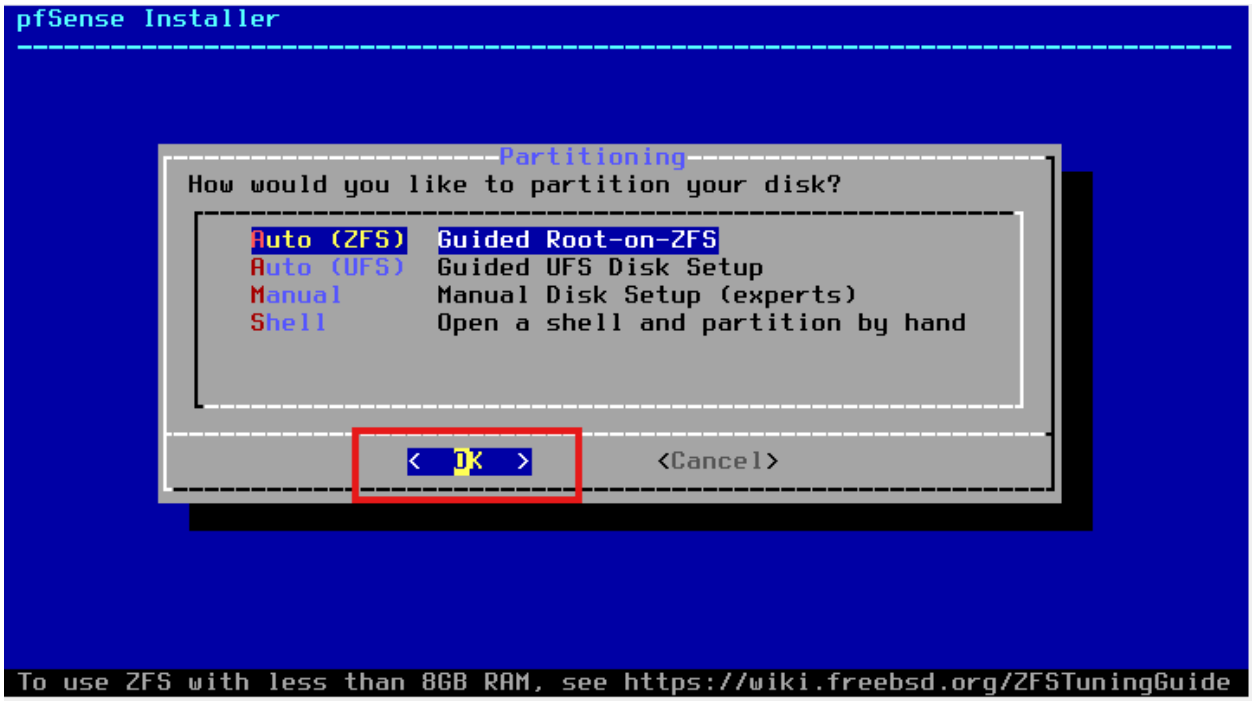
Your pfSense VM is now ready. Click *Start* to power it on.

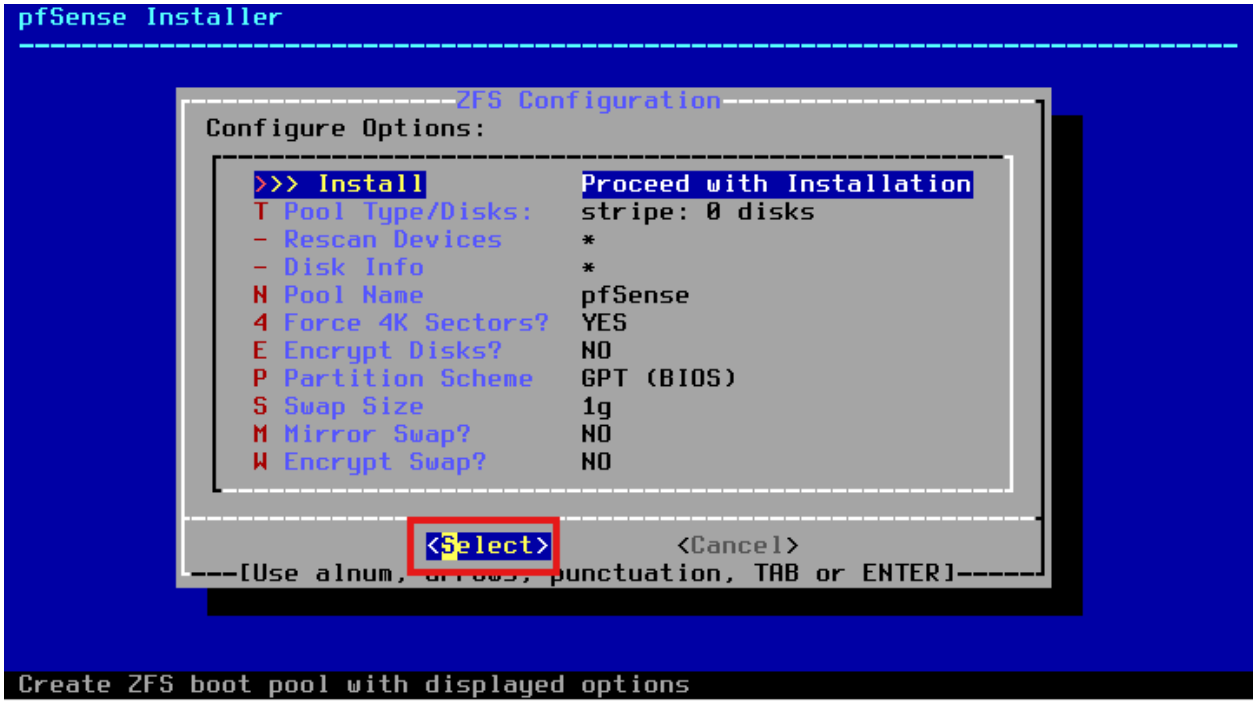


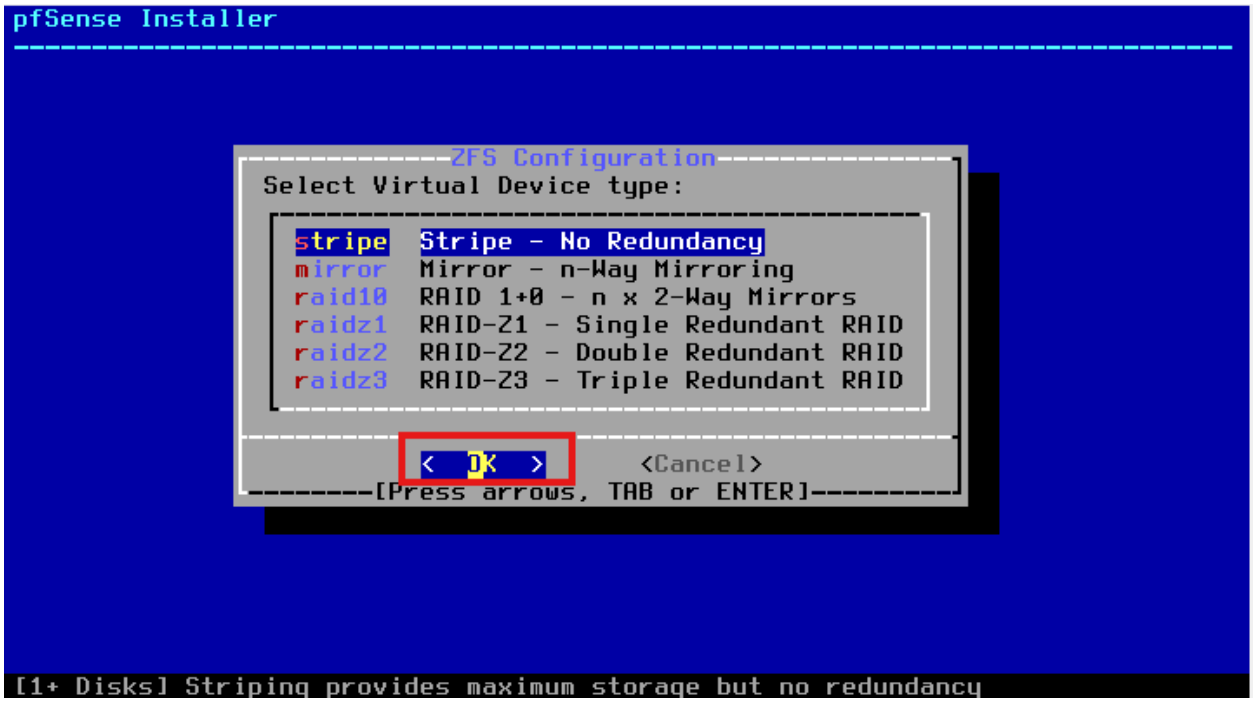
Follow the steps in the screenshots and do each step just like the pictures show.


pfsense [Running] - Oracle VirtualBox
File Machine View Input Devices Help
pfSense Installer
Copyright and distribution notice
Copyright and Trademark Notices.
Copyright 2004-2016. Electric Sheep Fencing, LLC ("ESF
All Rights Reserved.
Copyright 2014-2023. Rubicon Communications, LLC d/b/a Netgate
("Netgate").
All Rights Reserved.
All logos, text, and content of ESF and/or Netgate, including underlying
HTML code, designs, and graphics used and/or depicted herein are
protected under United States and international copyright and trademark
laws and treaties, and may not be used or reproduced without the prior
express written permission of ESF and/or Netgate.
"pfSense" is a registered trademark of ESF, exclusively licensed to
Netgate, and may not be used without the prior express written
permission of ESF and/or Netgate. All other trademarks shown herein are
26%
[Accept]
Mouse integration ...
Auto capture keyboard ...
Right Ctrl

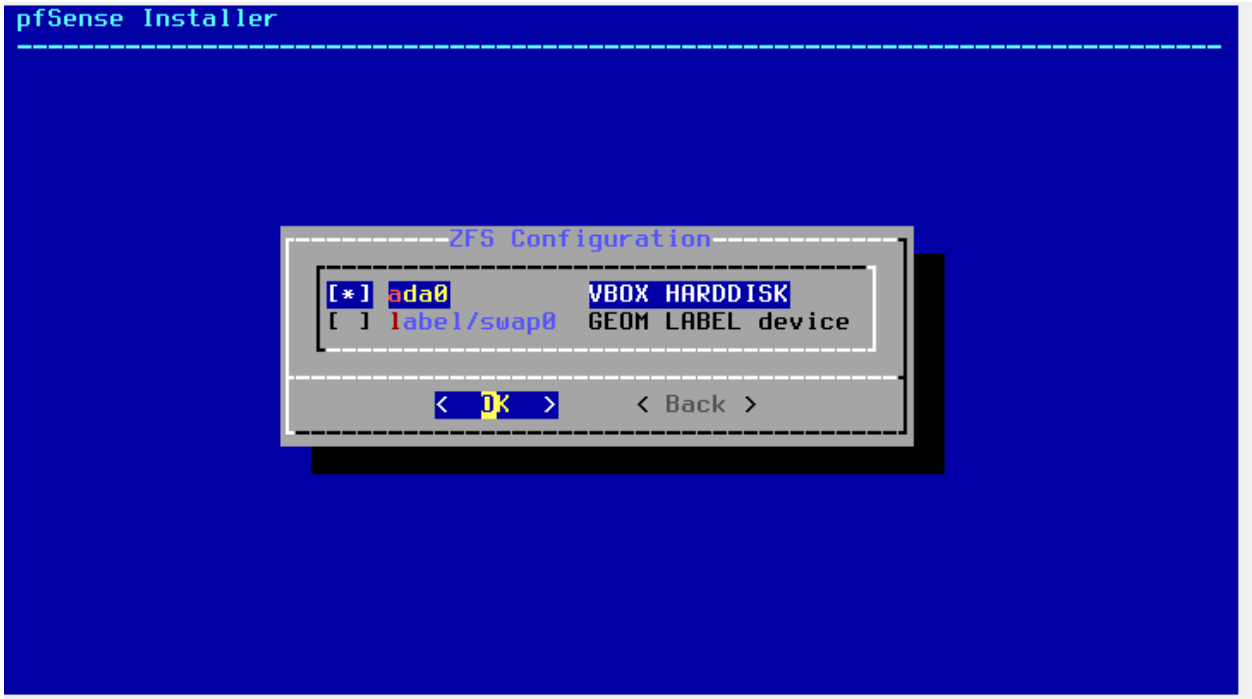

pfsense [Running] - Oracle VirtualBox
File Machine View Input Devices Help
pfSense Installer
Welcome
Welcome to pfSense!
Install Install pfSense
Rescue Shell Launch a shell for rescue operations
Recover config.xml Recover config.xml from a previous install
< OK >
<Cancel>
Mouse integration ...
Auto capture keyboard ...
Right Ctrl

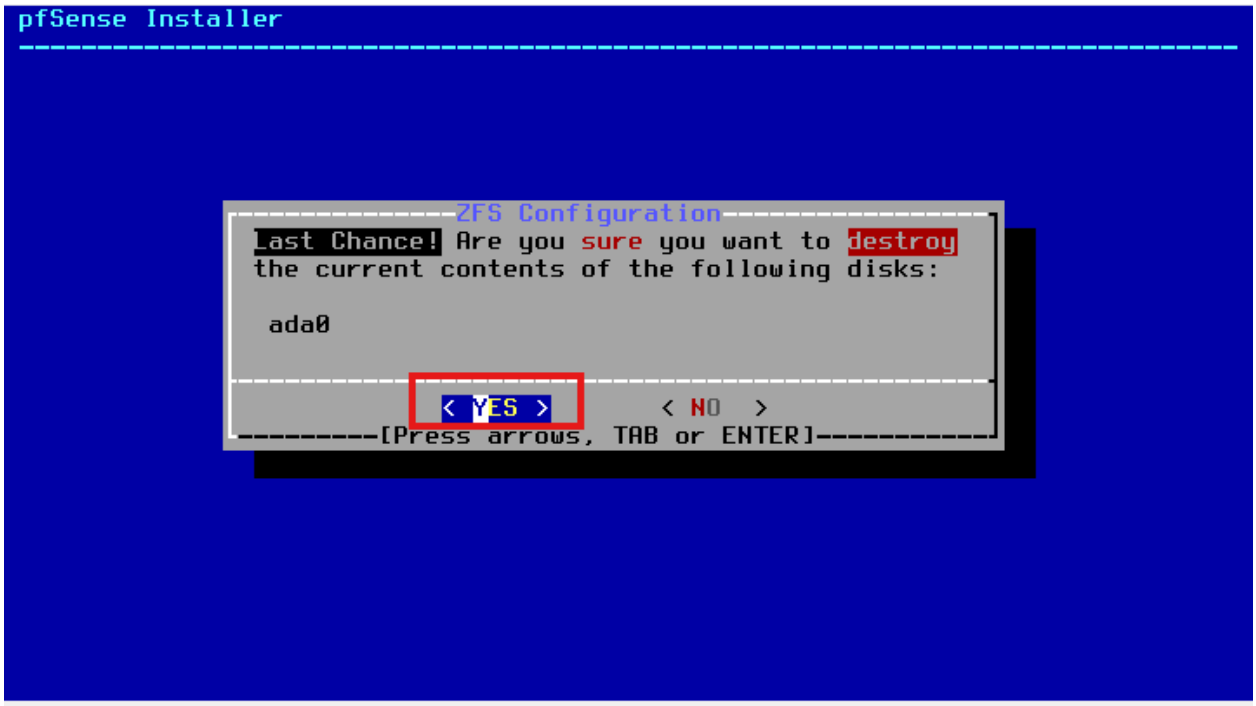

pfSense Installer
Partitioning
How would you like to partition your disk?
Auto (ZFS) Guided Root-on-ZFS
Auto (UFS) Guided UFS Disk Setup
Manual Manual Disk Setup (experts)
Shell Open a shell and partition by hand
< OK >
<Cancel>
To use ZFS with less than 8GB RAM, see https://wiki.freebsd.org/ZFSTuningGuide

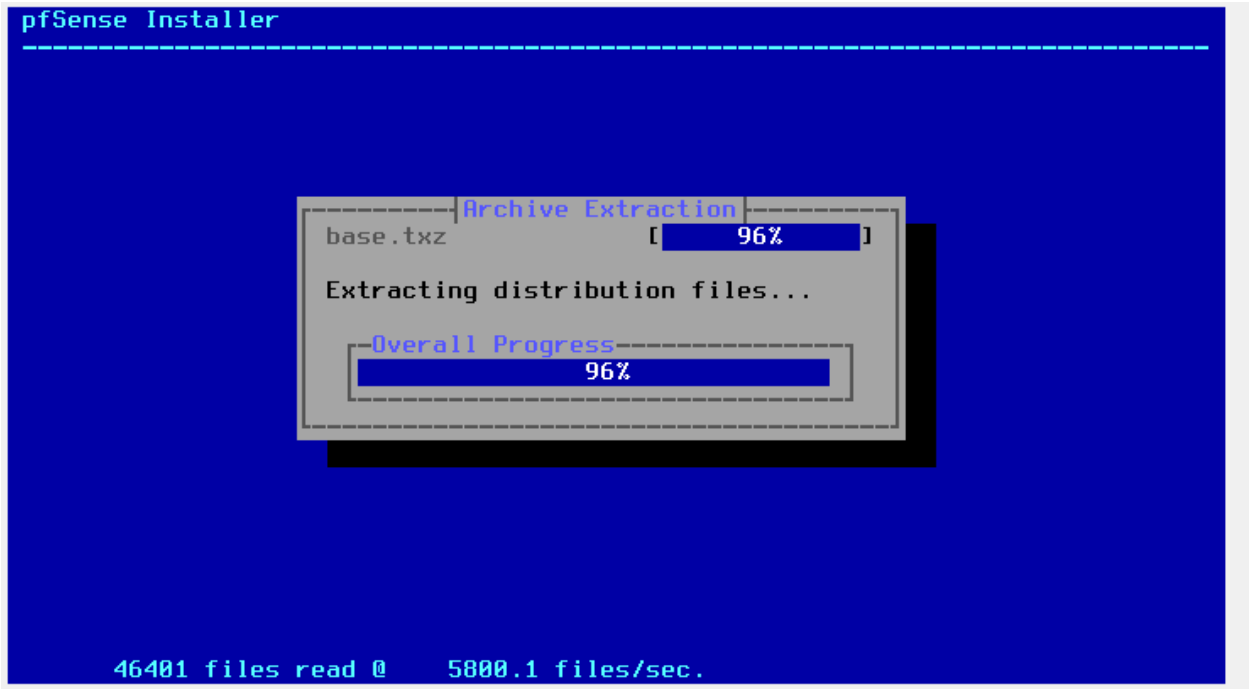

pfSense Installer
ZFS Configuration
Configure Options:
>>> Install Proceed with Installation
T Pool Type/Disks: stripe: 0 disks
- Rescan Devices *
- Disk Info *
N Pool Name pfSense
4 Force 4K Sectors? YES
E Encrypt Disks? NO
P Partition Scheme GPT (BIOS)
S Swap Size 1g
M Mirror Swap? NO
W Encrypt Swap? NO
<Select>
<Cancel>
Create ZFS boot pool with displayed options



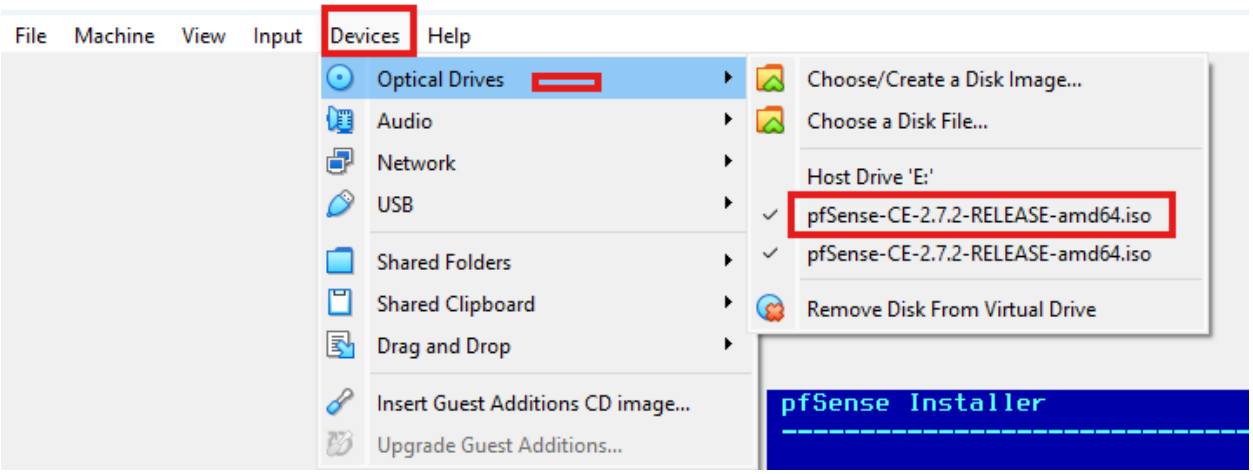
After pressing Enter, press Space.

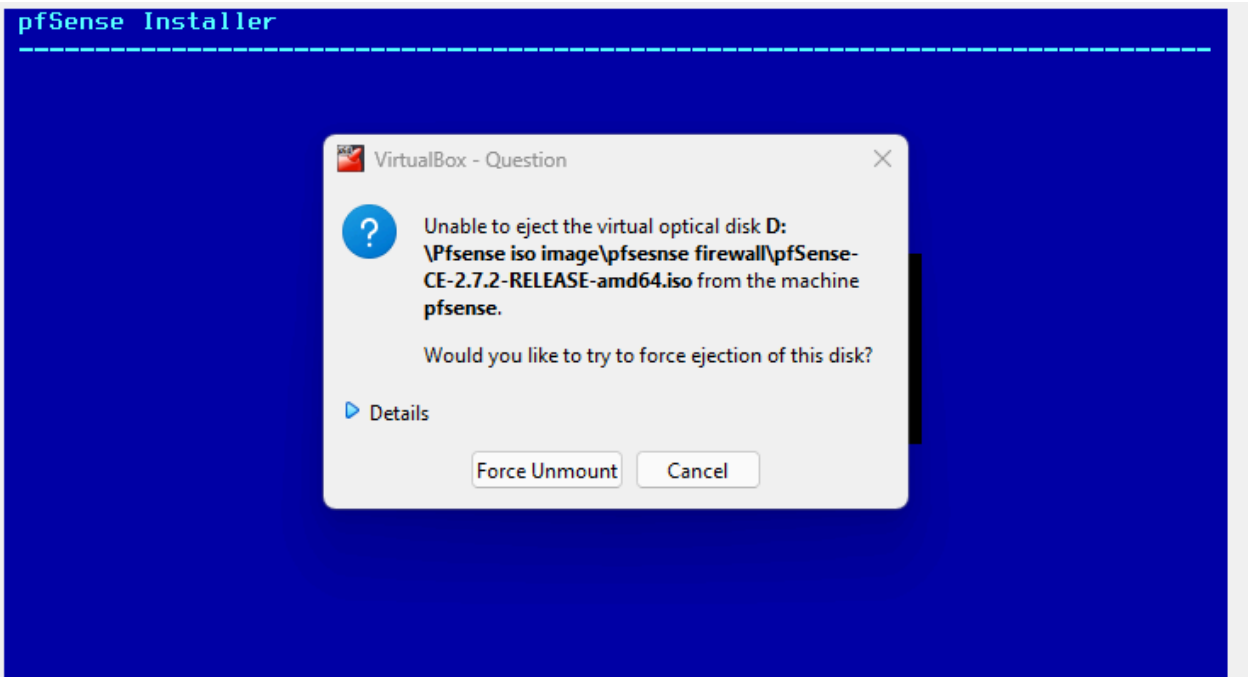




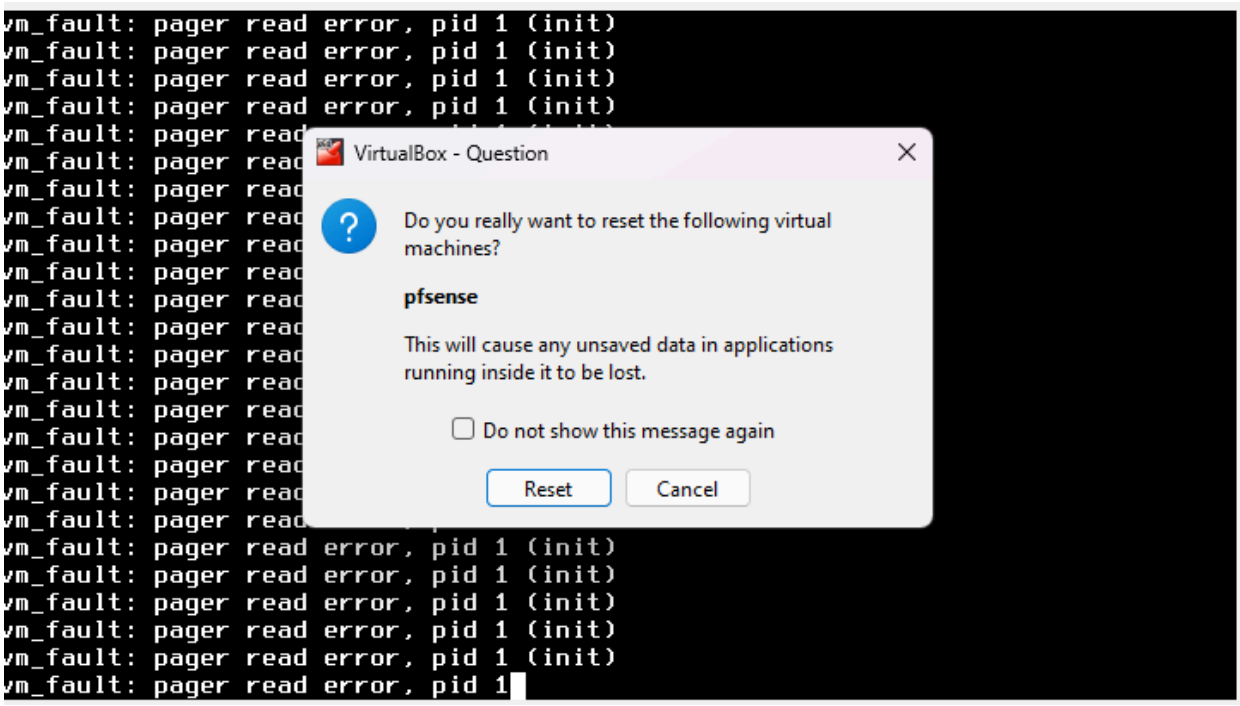


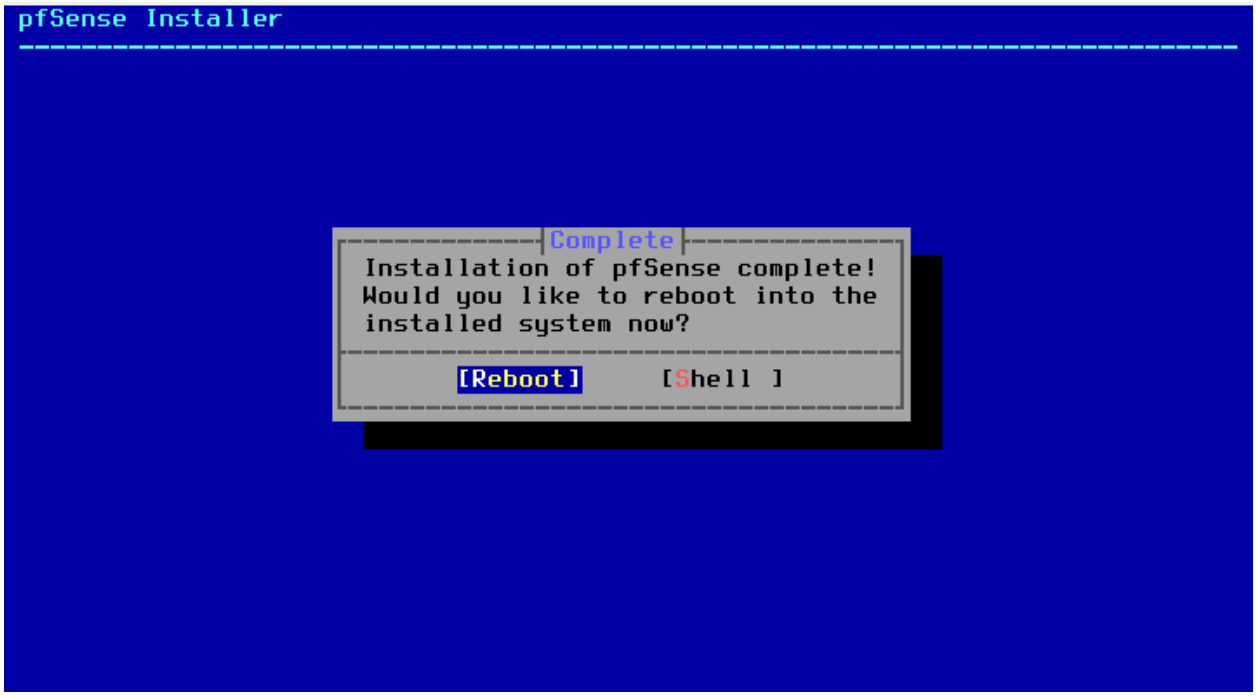
Click on the Device option, then select Optical Disk → ISO Image, and press Force Unmount and press enter.

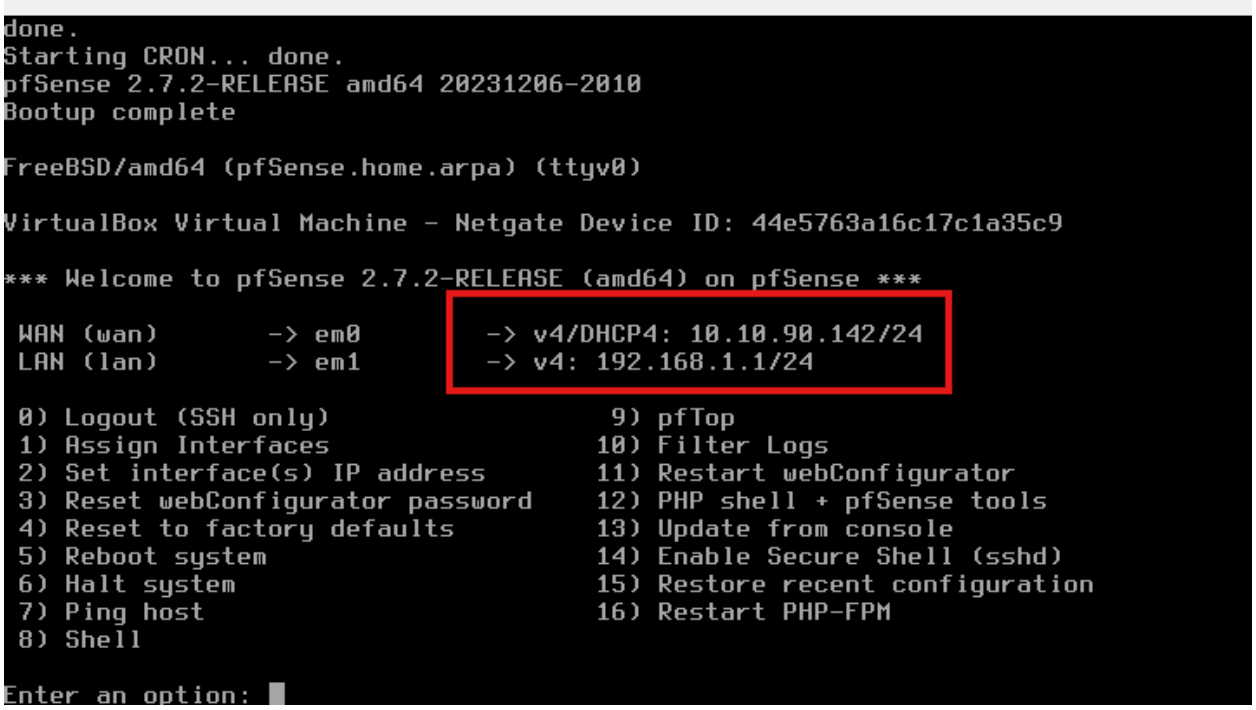

File
Machine
View
Input
Devices
Help
Optical Drives
Audio
Network
USB
Shared Folders
Shared Clipboard
Drag and Drop
Insert Guest Additions CD image...
Upgrade Guest Additions...
Choose/Create a Disk Image...
Choose a Disk File...
Host Drive 'E:'
pfSense-CE-2.7.2-RELEASE-amd64.iso
pfSense-CE-2.7.2-RELEASE-amd64.iso
Remove Disk From Virtual Drive
pfSense Installer


pfSense Installer
VirtualBox - Question
Unable to eject the virtual optical disk D:\Pfsense iso image\pfsesnse firewall\pfSense-CE-2.7.2-RELEASE-amd64.iso from the machine pfsense.
Would you like to try to force ejection of this disk?
Details
Force Unmount
Cancel

Ignore the errors and reset the machine.





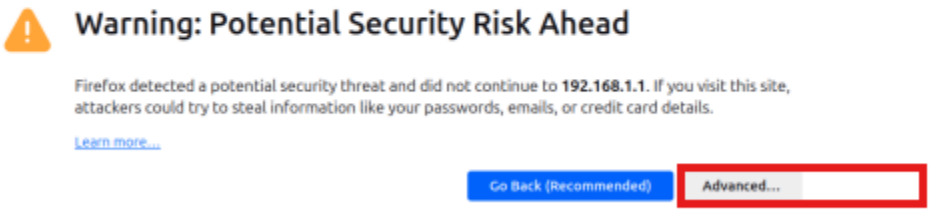


Next, switch to your Wazuh server or any Ubuntu system.

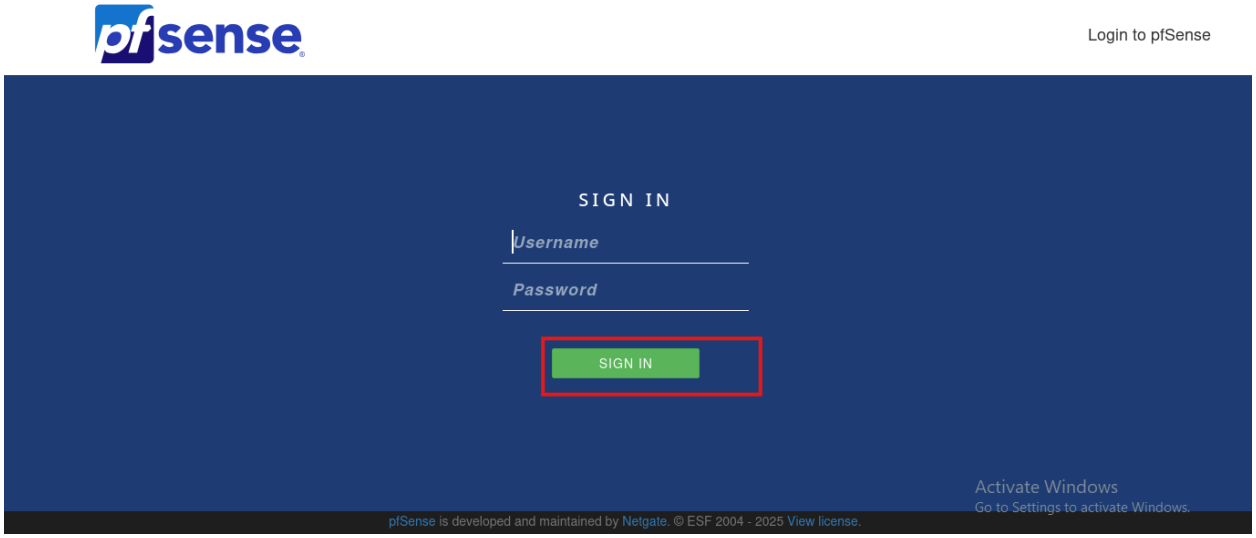
Note: Make sure this machine is also attached to the **Bridged Adapter**, the same way as your pfSense VM.

Once the machine is started, open a web browser and type the IP address displayed in the previous screenshot.

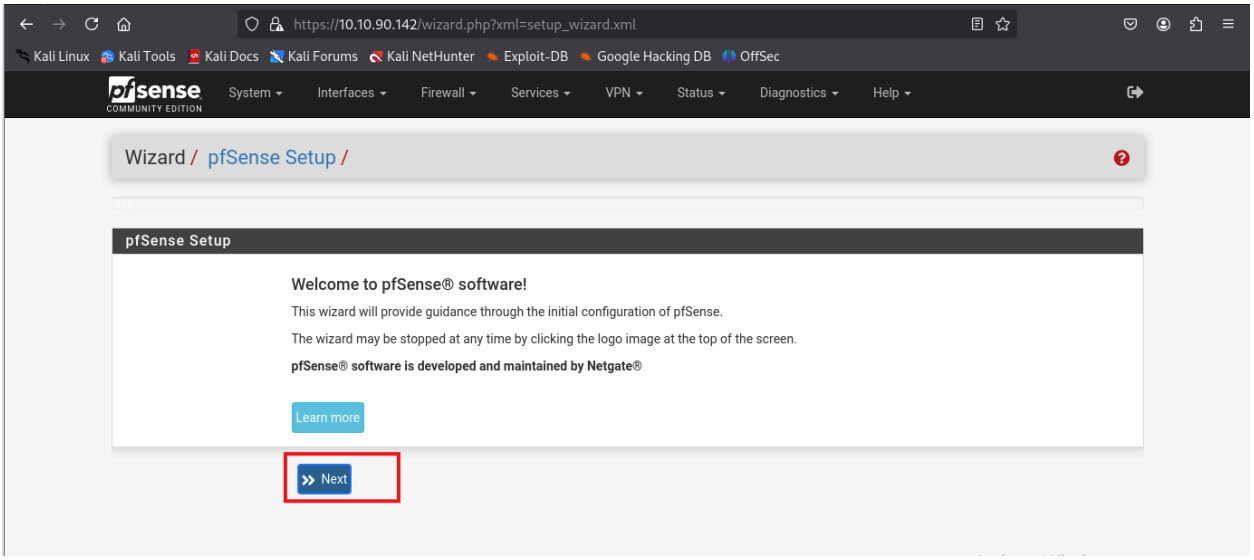
⚠️ Ensure that this system is on the **same bridged network** you are using to access the pfSense WebUI.

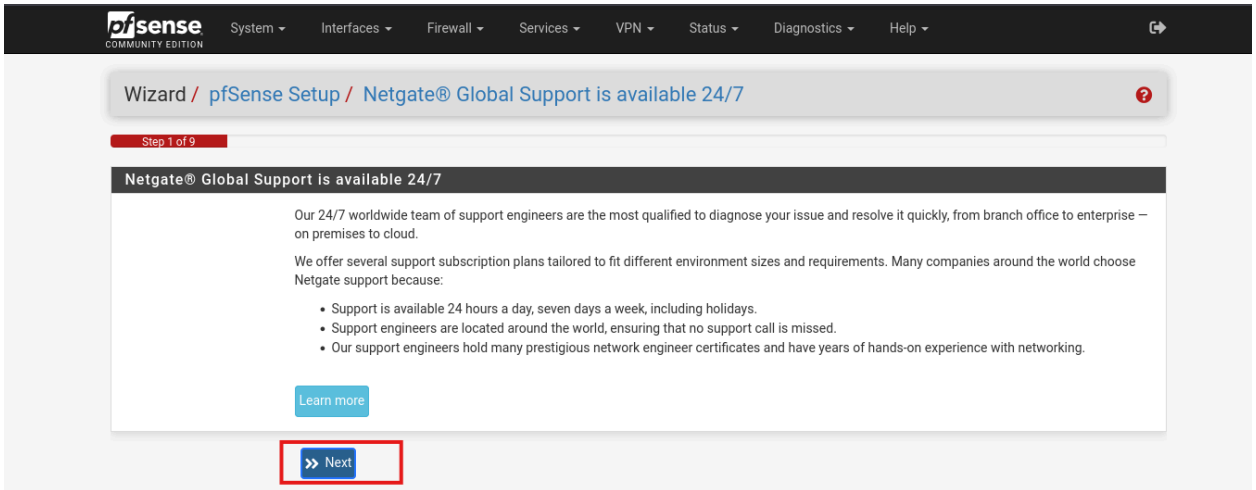


Click **Advanced** and confirm the security exception.
You will then be prompted for credentials:
Username: `admin`
Password: `pfsense`

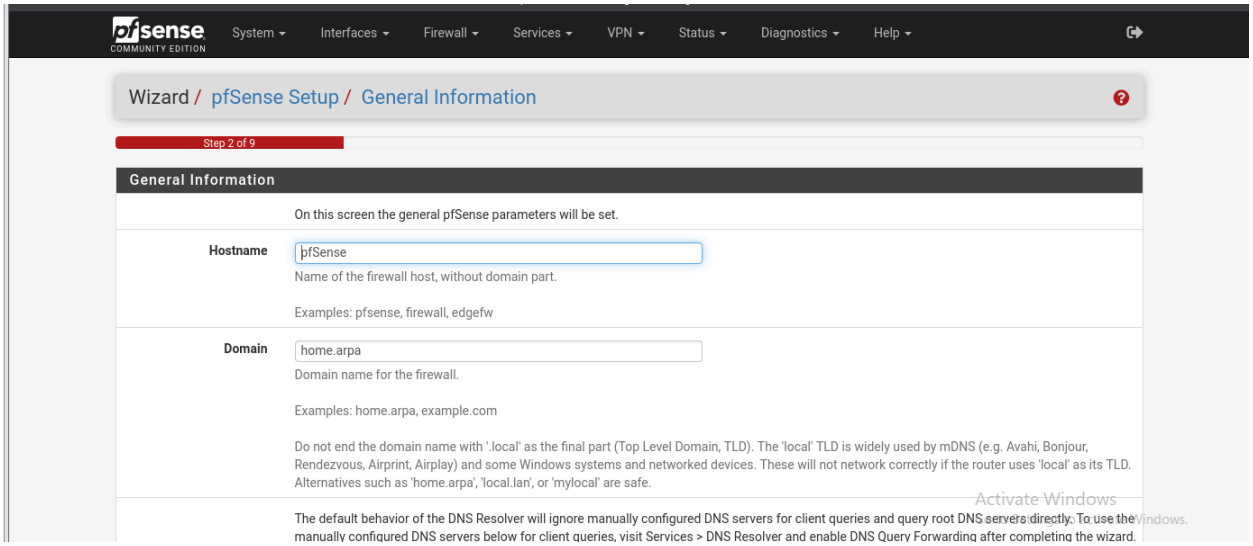


The following screen will appear. Please click **Next** to continue.

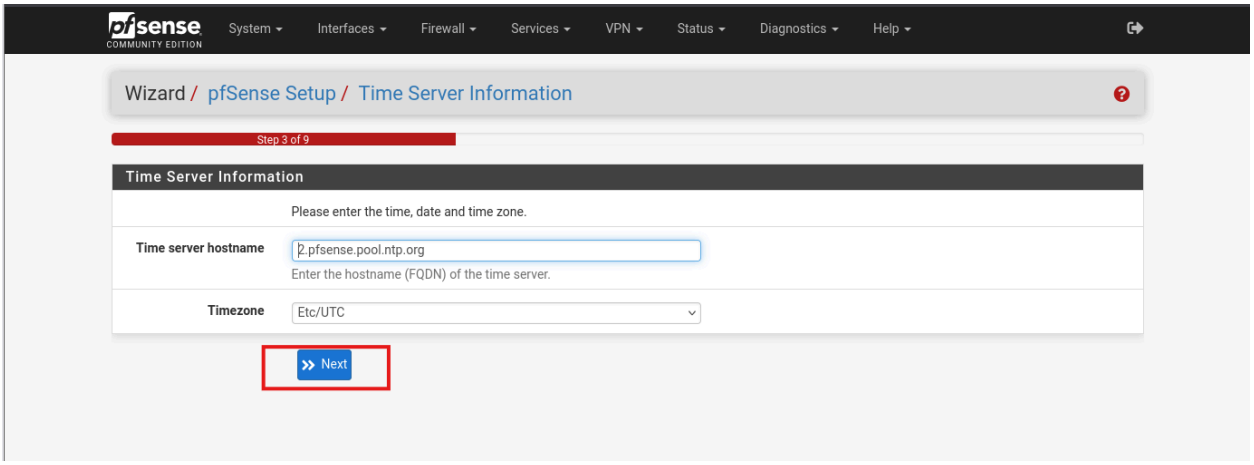




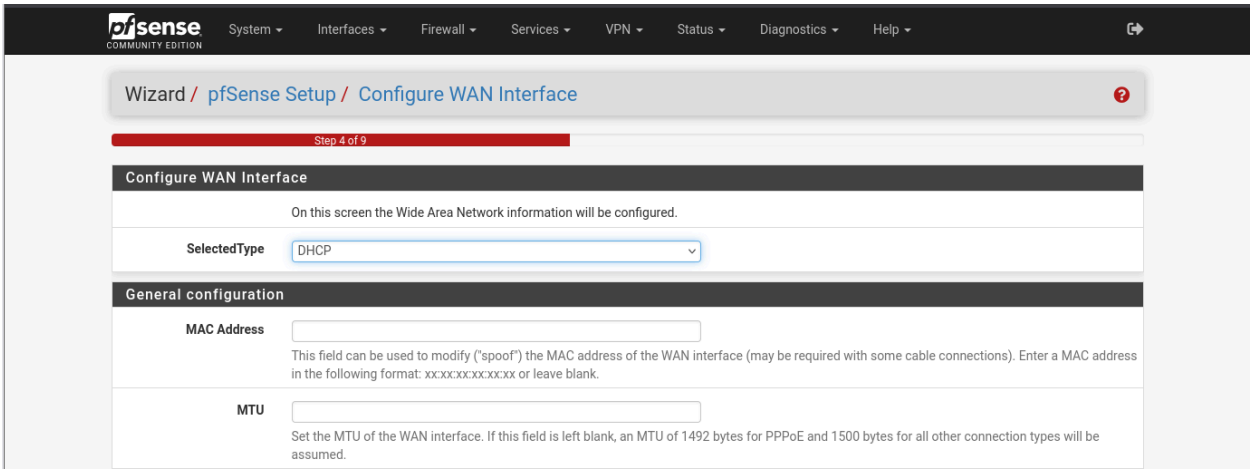
Click on next button



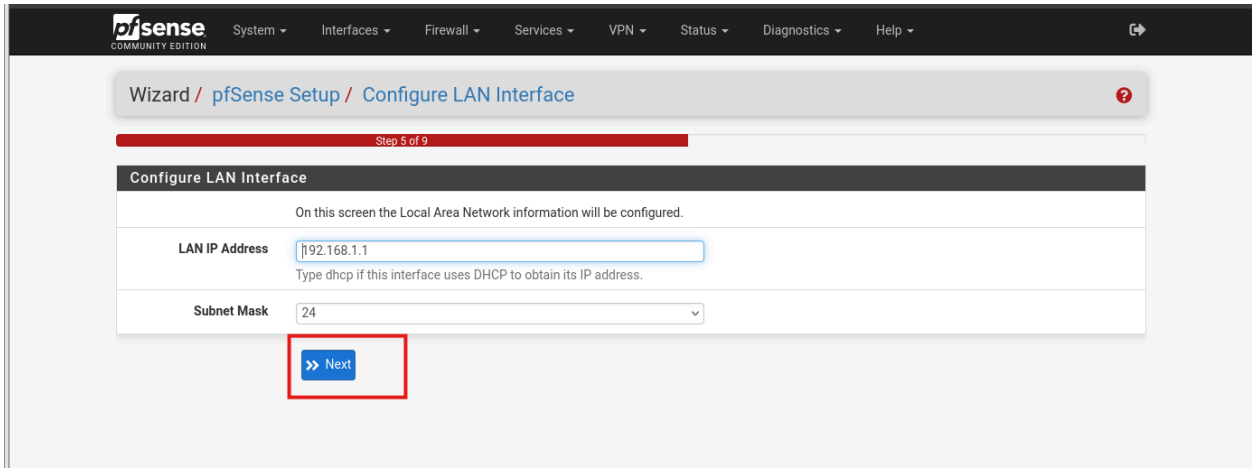
Scroll down and press the **Next** button.



Click on next button

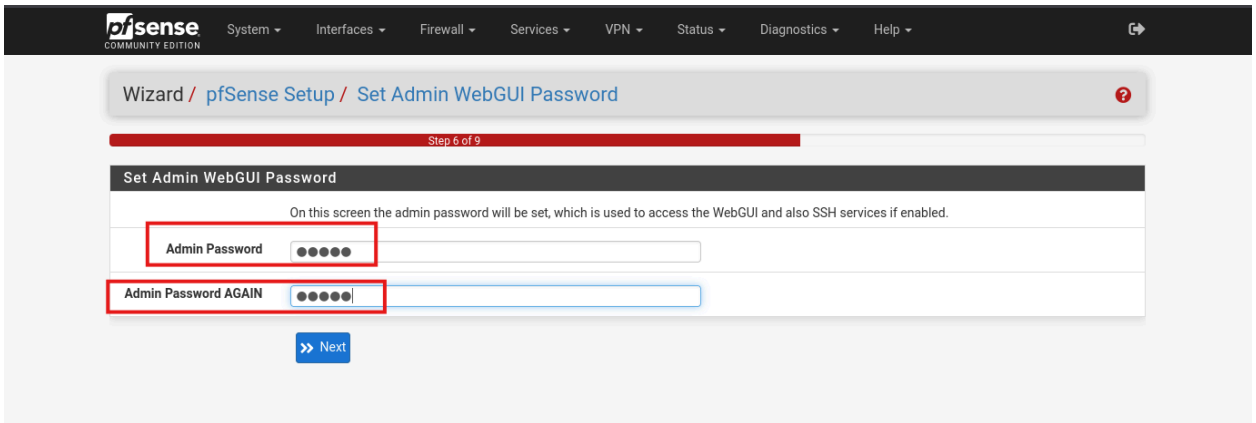


Scroll down and press the **Next** button.

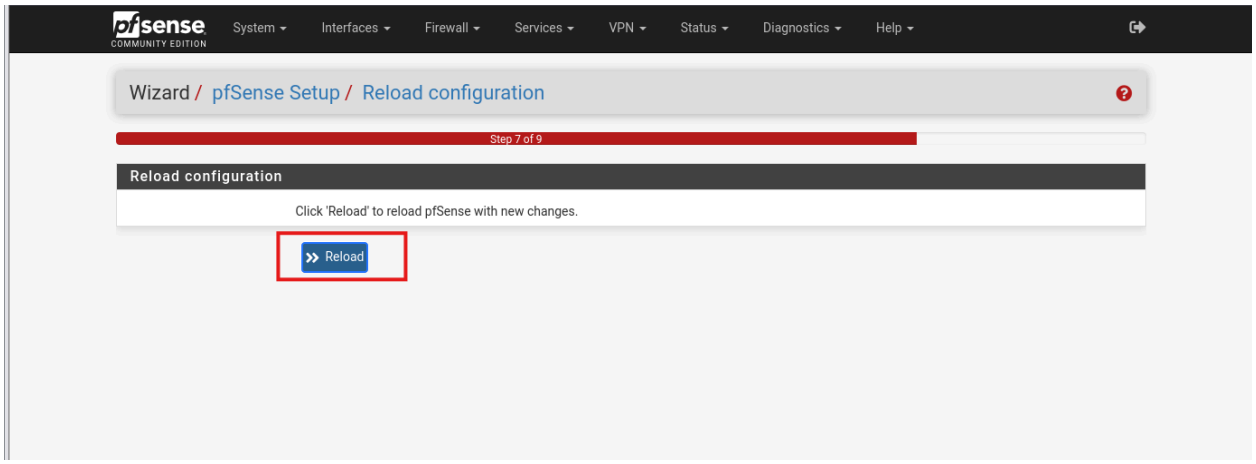


Click on next button

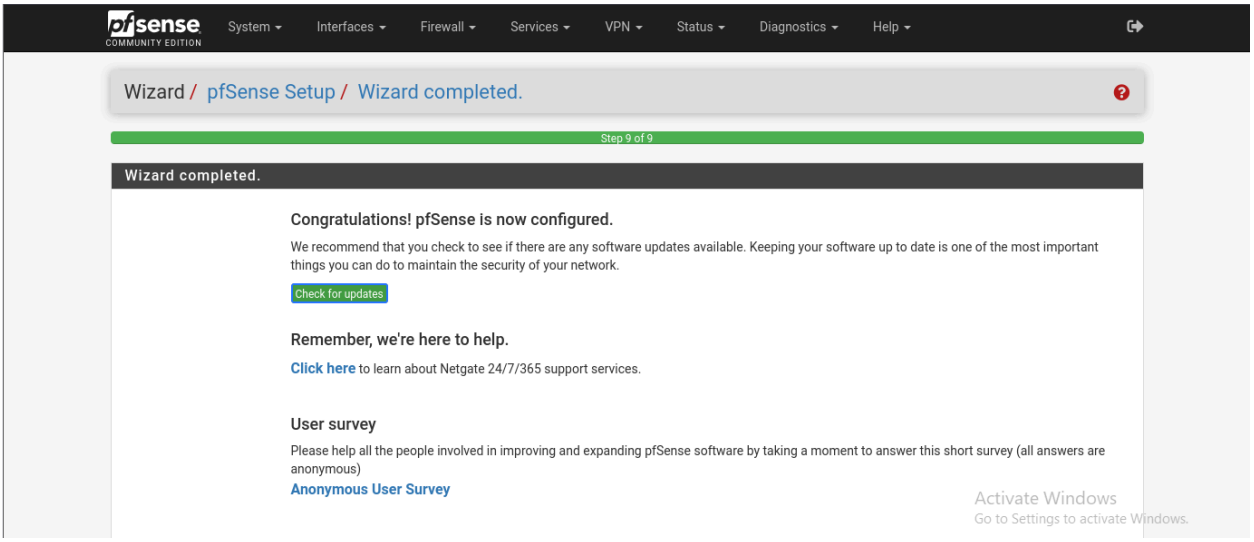
Set up your pfSense username and password, and proceed by clicking **Next**.

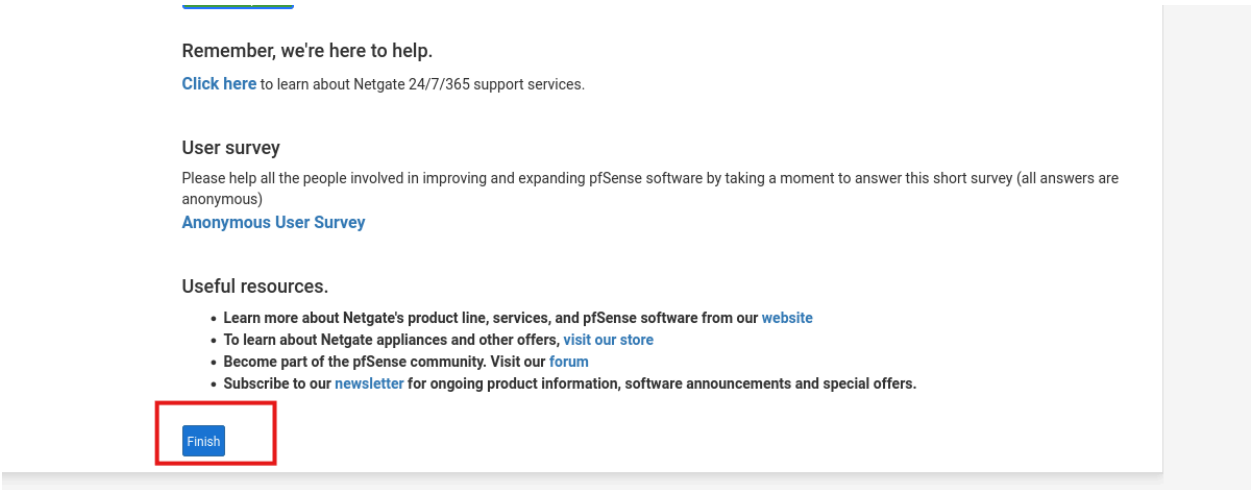


Click on next button



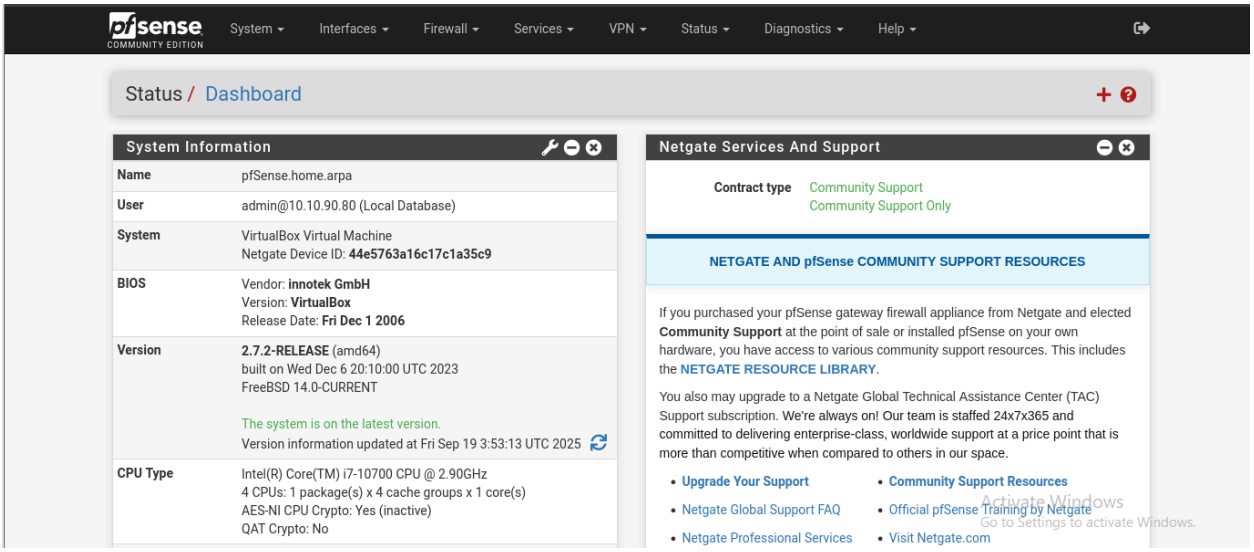
Please move to the bottom of the page and select the **Finish** button.





Now, please click on Accept button.

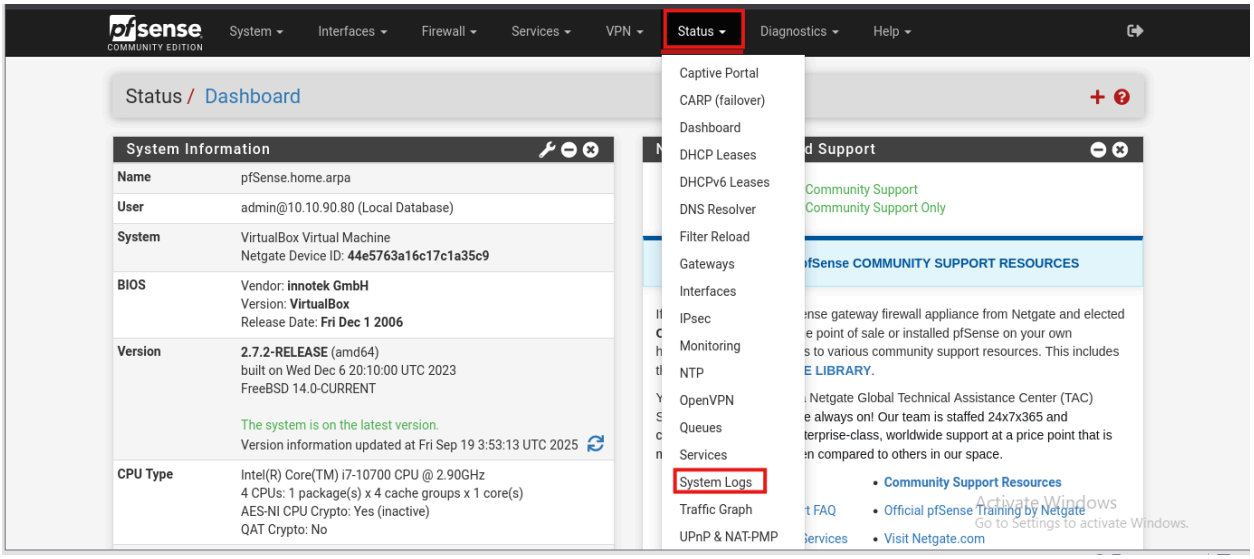
pfSense has been successfully configured and is ready for use.

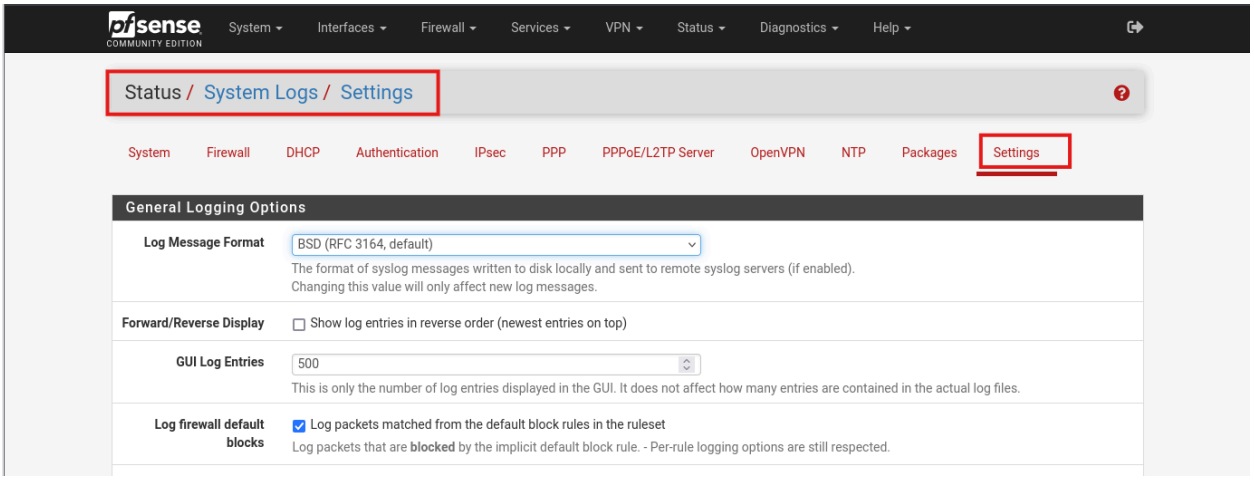


Integrating pfSense with the Wazuh Server

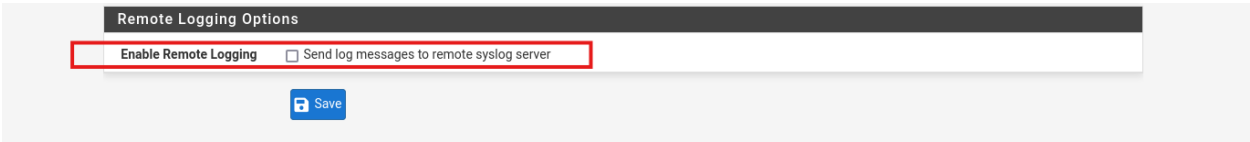
In this step, we will enable remote logging in pfSense to integrate it with the Wazuh server.

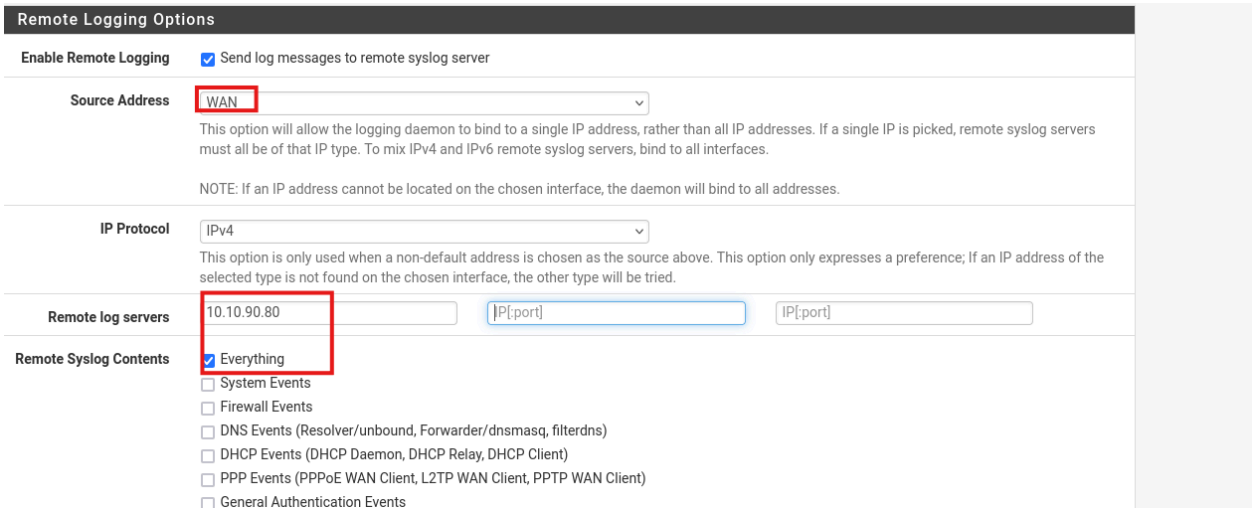
- Open the pfSense Web Interface.
- Navigate to **Status > System Logs > Settings**.



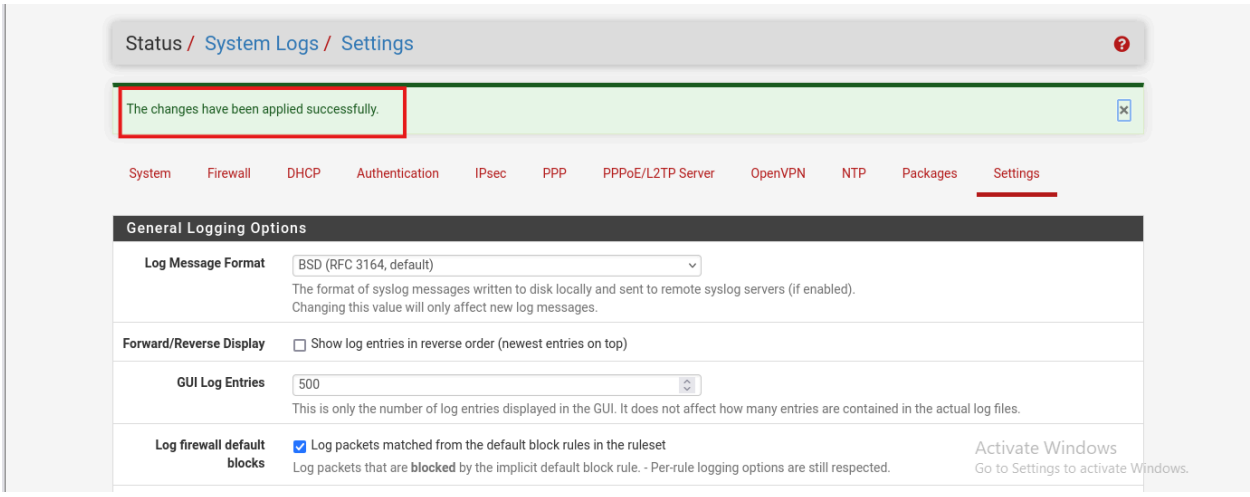


Scroll down to the bottom of the page, where you will find the “Enable Remote Logging” option. Enable it and configure the Wazuh Manager’s IP address with port 514 as the remote syslog server. 10.10.90.80:514



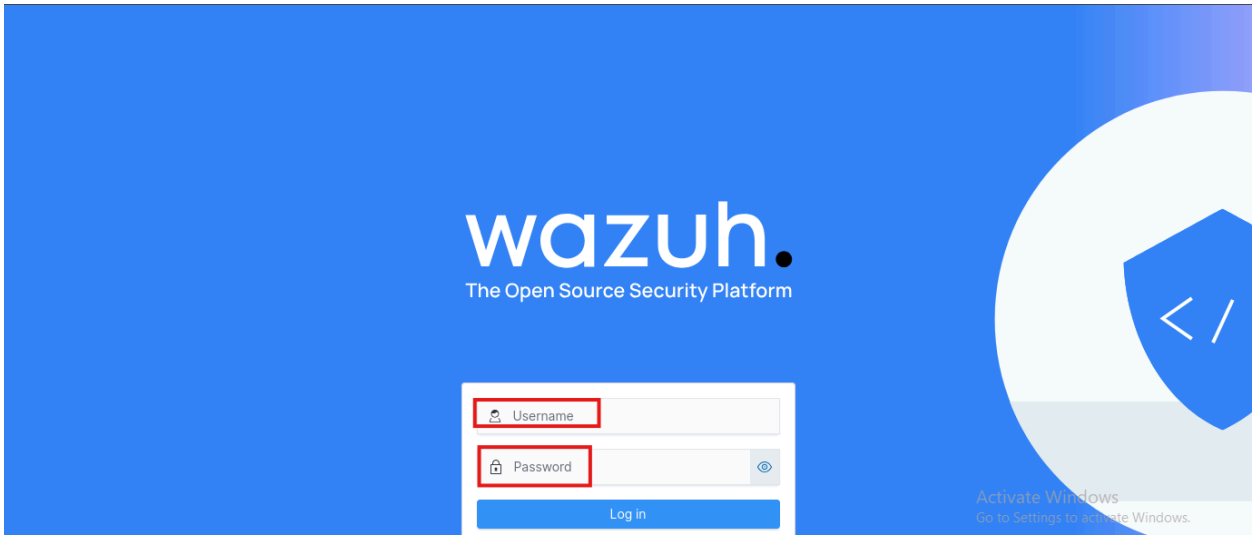


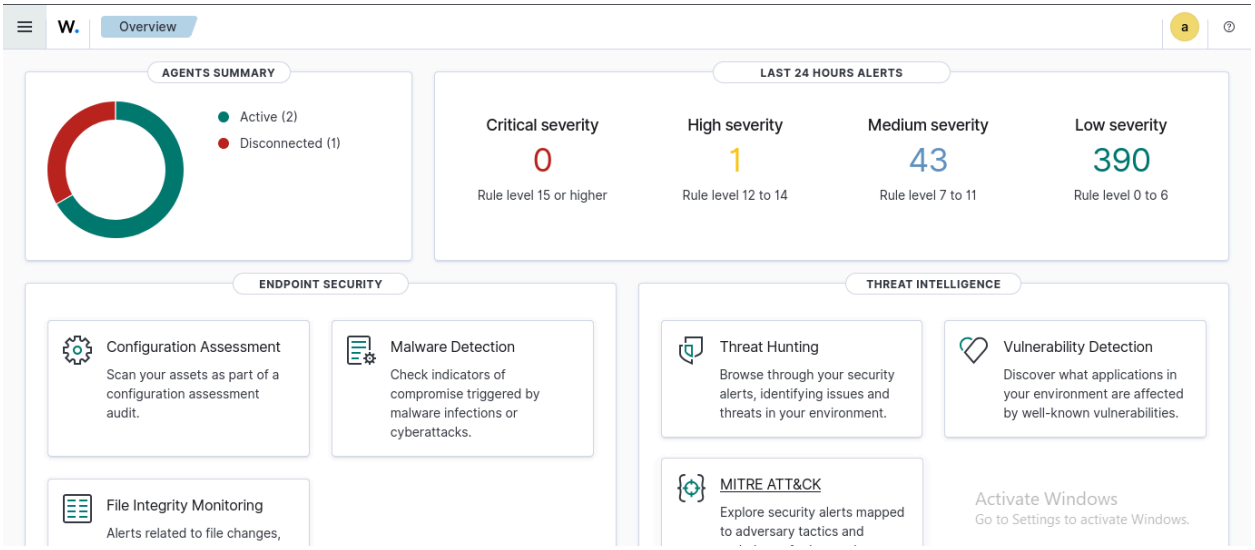
Click on **Save** and then **Apply Changes**.



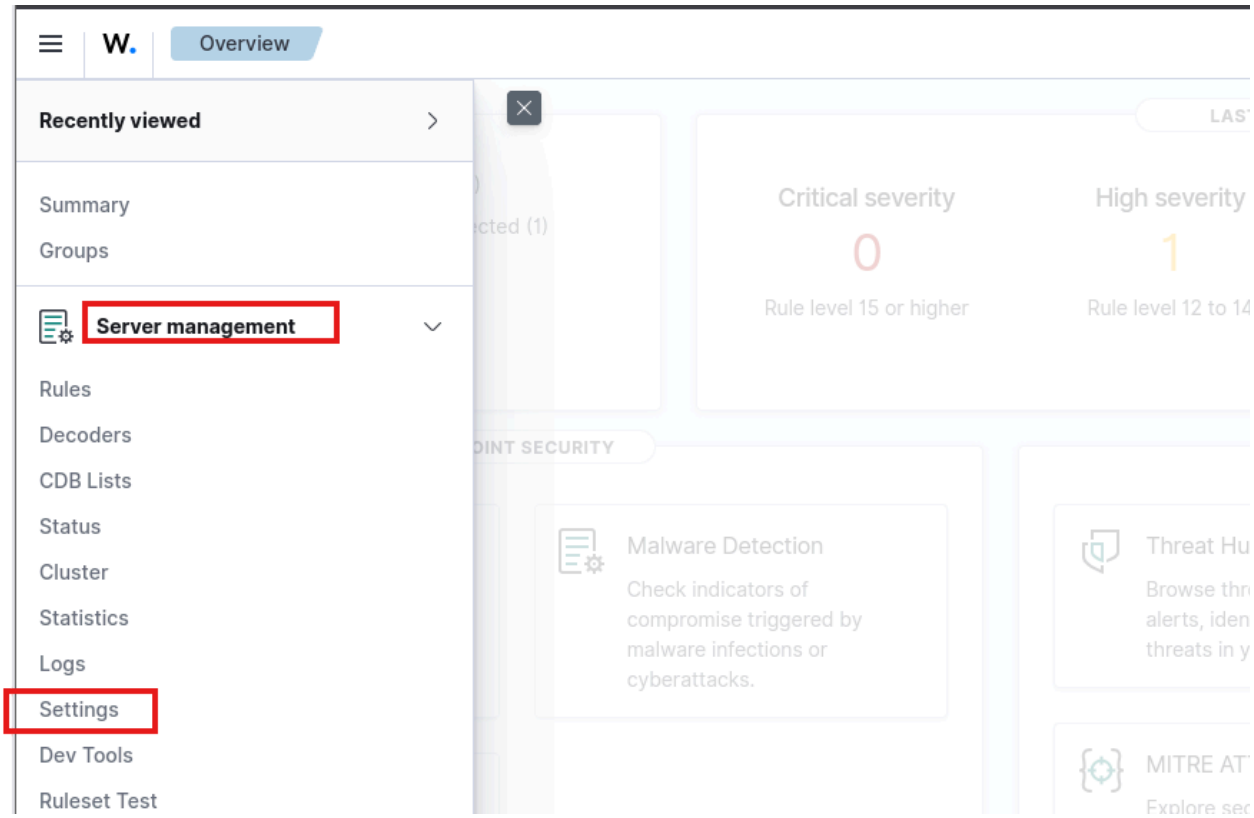
Custom Syslog Setup for UI Logs in Wazuh Server

To configure this, log in to your Wazuh server

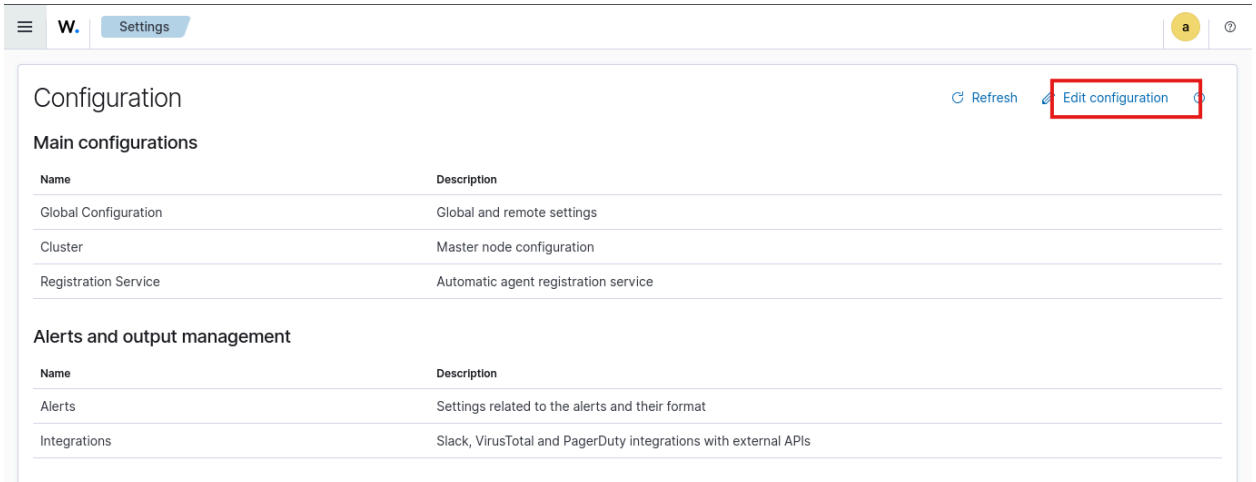




Navigate to **Server Management > Settings.**



Now, select the Edit Configuration option.



Edit the file **/var/ossec/etc/ossec.conf** on your Wazuh Manager.

🔹 **Important:** Replace the pfSense IP and Wazuh server IP with your actual IP addresses.

Inside the `<remote>` section, add the following configuration:

```
<!-- pfSense Firewall Integration -->

<remote>

<connection>syslog</connection>

<port>514</port>

<protocol>udp</protocol>
```

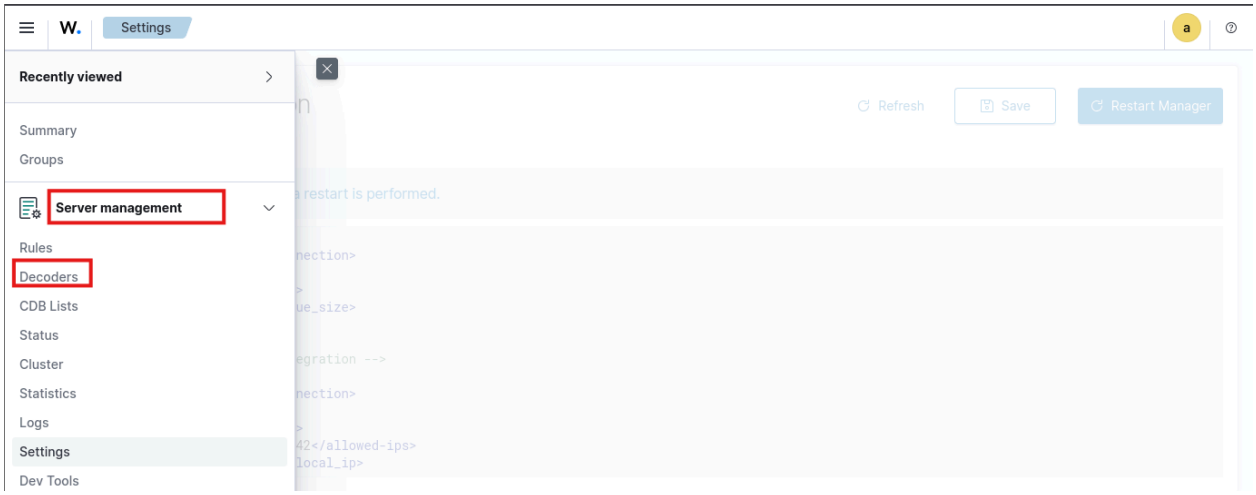
<allowed-ips>Pfsense WAN IP/24</allowed-ips>

<local_ip>Wazuh IP</local_ip>

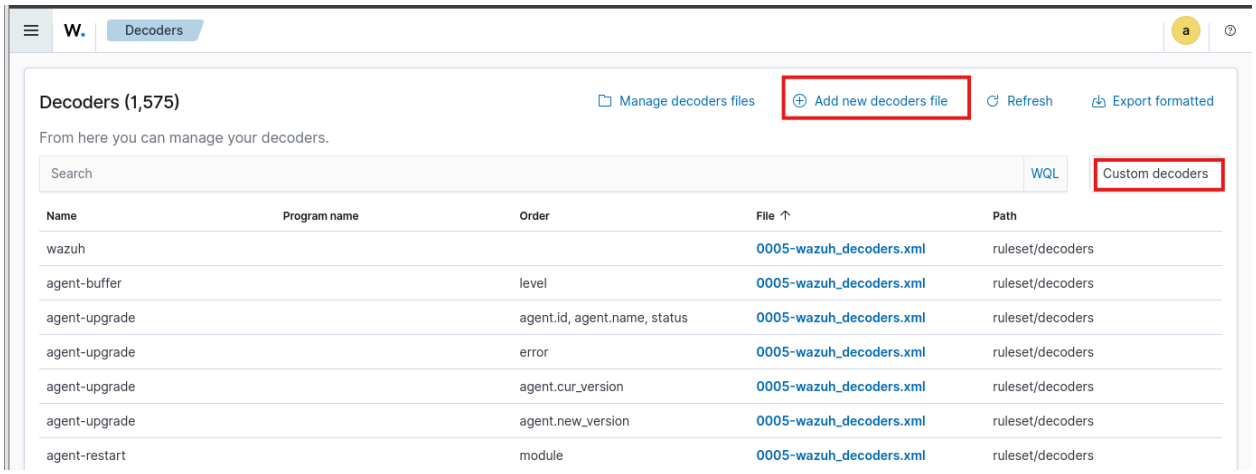
</remote>

Click on the save button and restart manager.

Now, go back to **Server Management** and add the **Custom Decoders and Rules** to enable proper UI log parsing.



Please select the **Custom Decoders** option from the menu.



Click on **Add new decoder file** and enter the name:

Pfsense-custom-decoder.xml

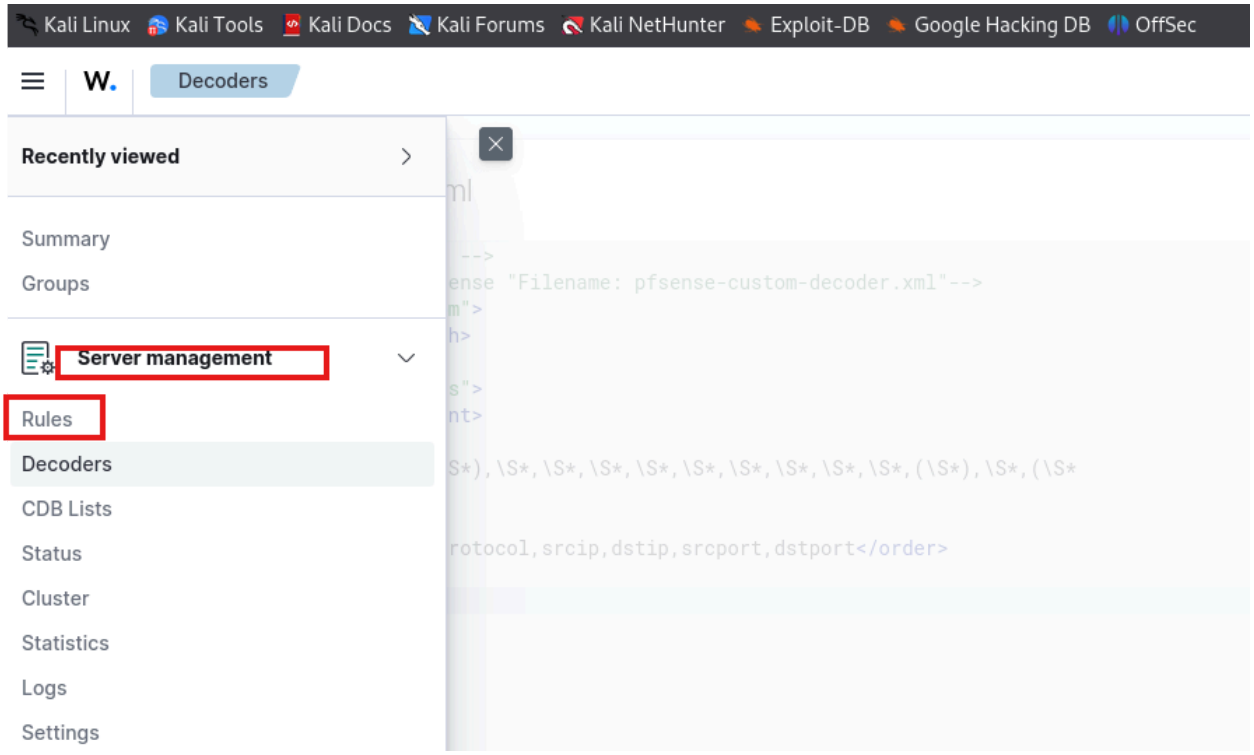
Custom decoder: Create pfsense-custom-decoder.xml in /var/ossec/etc/decoders/.

```
<!-- add new decoder for pfsense "Filename: pfsense-custom-decoder.xml"-->

<decoder name="pfsense-custom">

<prematch>filterlog</prematch>

</decoder>

<decoder name="pfsense-fields">

<parent>pfsense-custom</parent>

<regex>^(\w+)[\d+]:

\S*,\S*,\S*,(\S*),\S*,\S*,(\S*),\S*,\S*,\S*,\S*,\S*,\S*,\S*,\S*,\S*,(\S*),\S*,(\S*

),(\S*),(\d*),(\d

),\S</regex>
```

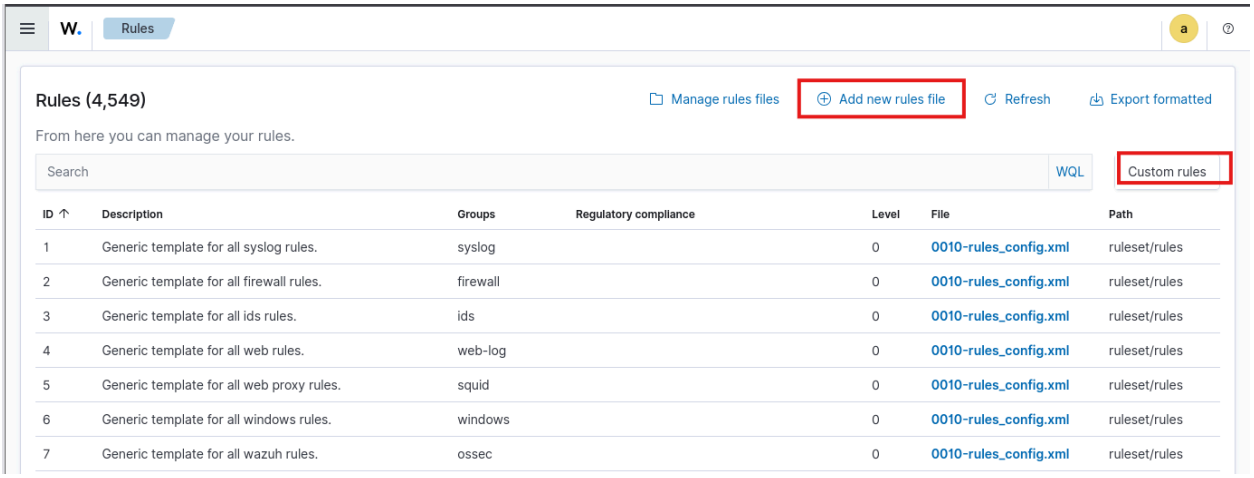
```
<order>logsource,id,action,protocol,srcip,dstip,srcport,dstport<
/order>

</decoder>
```

Click on save button.

Now add the custom rules.



Select the custom rules and add new rules files.



Click on **Add new rules file** and enter the name:

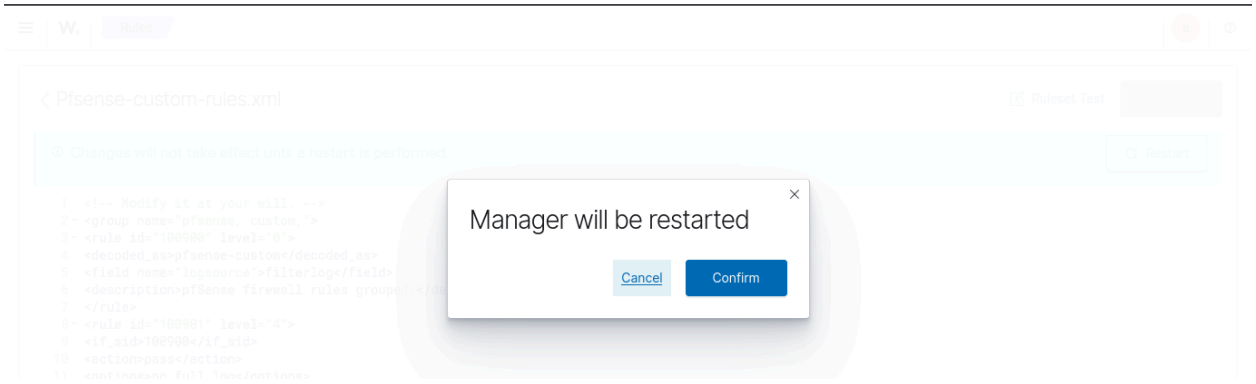
Pfsense-custom-rules.xml

```
<group name="pfsense, custom,">
  <rule id="100900" level="0">
    <decoded_as>pfsense-custom</decoded_as>
    <field name="logsource">filterlog</field>
    <description>pfSense firewall rules grouped.</description>
  </rule>


  <rule id="100901" level="4">
    <if_sid>100900</if_sid>
    <action>pass</action>
    <options>no_full_log</options>
    <description>pfSense firewall allow event.</description>
```

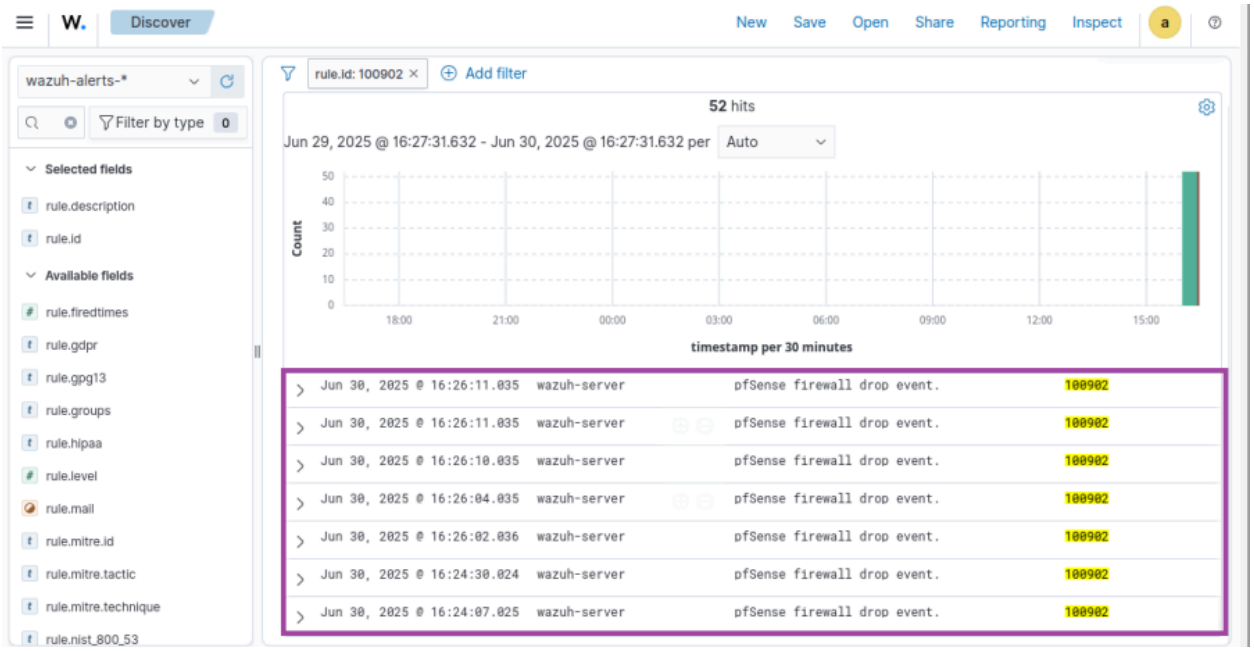
```
<group>firewall_allow,pci_dss_1.4,gpg13_4.12,hipaa_164.312.a.1,n
ist_800_53_SC.7,tsc_CC6.7,tsc_CC6.8,</group>

  </rule>


  <rule id="100902" level="5">

    <if_sid>100900</if_sid>

    <action>block</action>

    <options>no_full_log</options>

    <description>pfSense firewall drop event.</description>


<group>firewall_block,pci_dss_1.4,gpg13_4.12,hipaa_164.312.a.1,n
ist_800_53_SC.7,tsc_CC6.7,tsc_CC6.8,</group>

  </rule>


  <rule id="100903" level="10" frequency="18" timeframe="45"
ignore="240">

    <if_matched_sid>100902</if_matched_sid>

    <same_source_ip />

    <description>Multiple pfSense firewall blocks events from
same source.</description>

    <mitre>

      <id>T1110</id>

    </mitre>
```

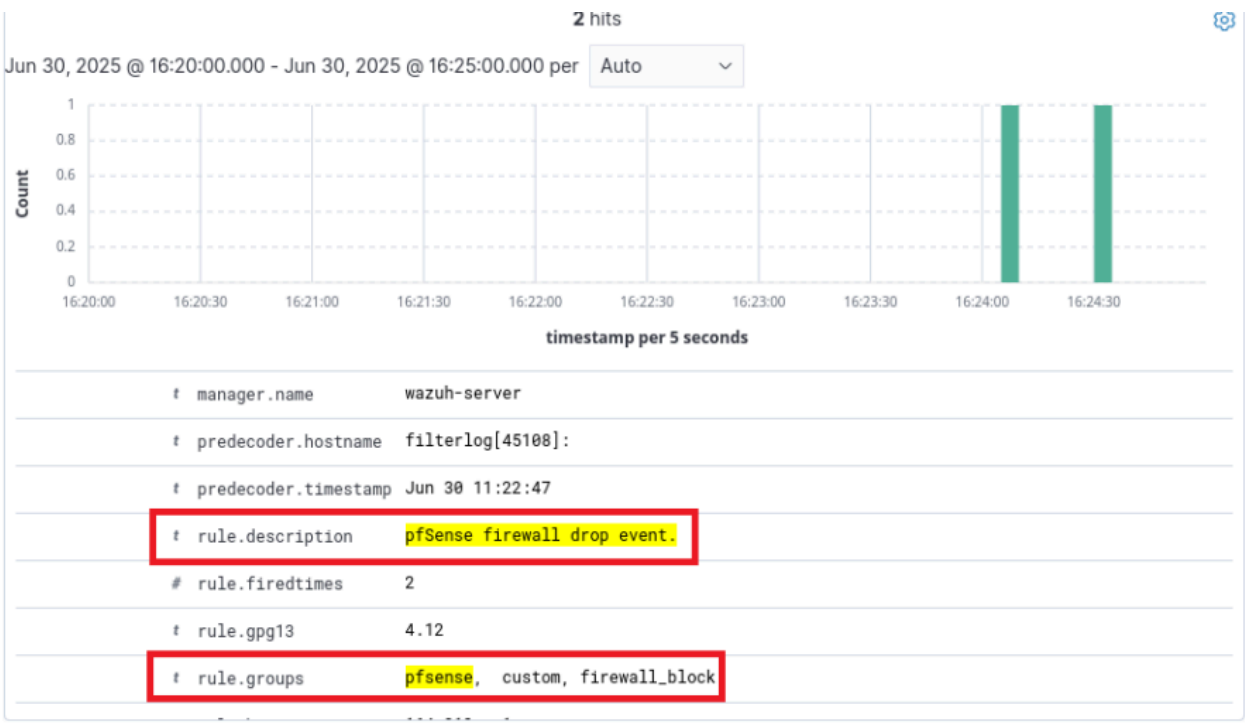
```
<group>multiple_blocks,pci_dss_1.4,pci_dss_10.6.1,gpg13_4.12,hipaa_164.312.a.1,hipaa_164.312.b,nist_800_53_SC.7,nist_800_53_AU.6,tsc_CC6.7,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,</group>

  </rule>

</group>
```

Click save button and restart the wazuh manager.

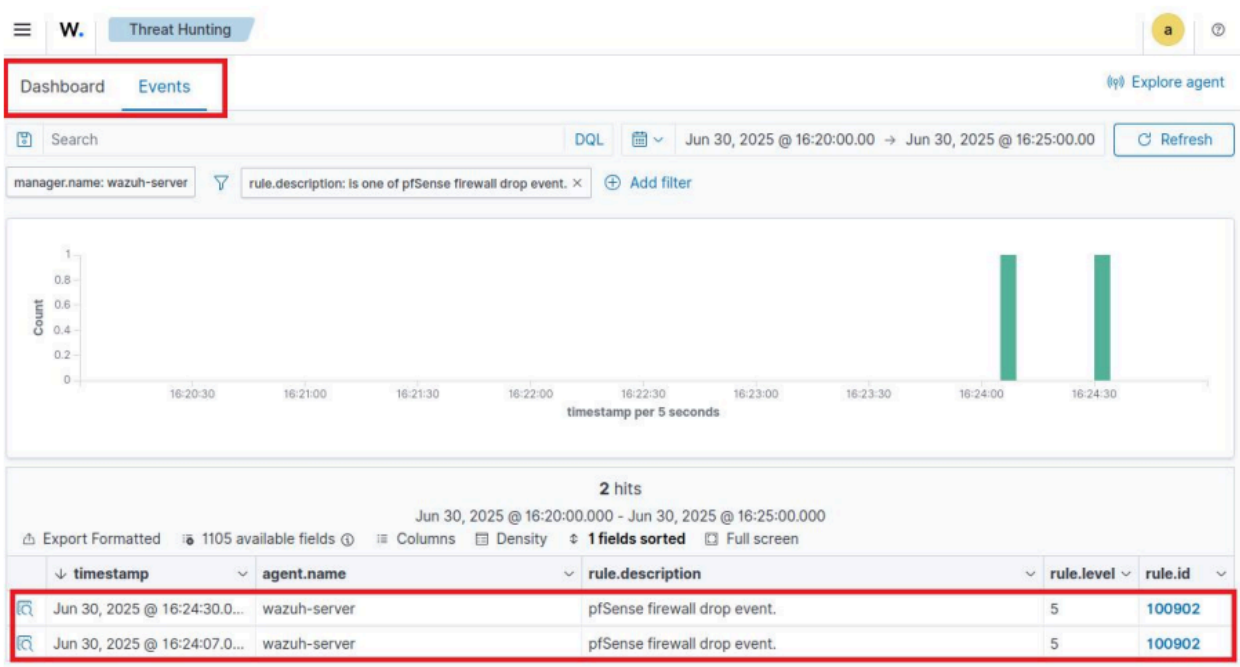


Monitor your log in wazuh-server





wazuh-alerts-*
Filter by type
Selected fields
_source
Popular fields
rule.description
rule.id
timestamp
Available fields
_index
agent.id
agent.name
data.action
data.command
data.dstip
data.dstport
rule.id: is one of 100902
rule.description: pfSense firewall drop event.
Add filter
2 hits
Jun 30, 2025 @ 16:20:00.000 - Jun 30, 2025 @ 16:25:00.000 per Auto
Count
timestamp per 5 seconds
predecoder.timestamp Jun 30 11:22:24
rule.description pfSense firewall drop event.
rule.firedtimes 1
rule.gpg13 4.12
rule.groups pfsense, custom, firewall_block
rule.hipaa 164.312.a.1
rule.id 100902


Threat Hunting
Dashboard
Events
Explore agent
Search
DQL
Jun 30, 2025 @ 16:20:00.00 → Jun 30, 2025 @ 16:25:00.00
Refresh
manager.name: wazuh-server
rule.description: is one of pfSense firewall drop event.
Add filter
Count
timestamp per 5 seconds
2 hits
Jun 30, 2025 @ 16:20:00.000 - Jun 30, 2025 @ 16:25:00.000
Export Formatted
1105 available fields
Columns
Density
1 fields sorted
Full screen
timestamp
agent.name
rule.description
rule.level
rule.id
Jun 30, 2025 @ 16:24:30.0...
wazuh-server
pfSense firewall drop event.
5
100902
Jun 30, 2025 @ 16:24:07.0...
wazuh-server
pfSense firewall drop event.
5
100902

Summary:

We have successfully set up the integration between **pfSense and Wazuh** to strengthen network security monitoring. With this setup, Wazuh can now gather and process logs from pfSense, which helps in identifying unusual activities, firewall alerts, and irregularities in network traffic. The documentation we prepared covers all required steps, configurations, and screenshots so that others can follow along and implement the integration without difficulty.

2. Firewall Configuration and Rule Testing in Home Lab

A. Restrict inbound and outbound traffic from specific countries (for example, China and Russia).